

CLIM1001 Overall Lessons Content wk4-6

Lesson4:

1 - Anthropogenic climate forcing

Introduction

In earlier lessons you learned why the natural greenhouse effect is critical for life on Earth. Now we'll investigate how some human activities can affect the Earth's radiative equilibrium, including altering the strength of the greenhouse effect.

Before we begin, watch the video below. See if you can pick anything in the depiction of the science that isn't quite right...

Aerosol emissions

In the [greenhouse effect lesson](#), you learned how aerosols affect the Earth's radiation balance by scattering incoming [shortwave](#) energy and changing the [albedo](#) of clouds, generally having a cooling effect.

Many [anthropogenic](#) activities release aerosols in addition to natural sources. Sulphate aerosols are a significant by-product of industrial processes (figure below right), while land clearing and deforestation release large amounts of [black carbon](#) (particulates from the burning of organic matter) and [organic carbon](#) into the atmosphere. Most of these aerosols act just as natural aerosols do, cooling rather than warming the planet - the more aerosols in the air, the less [shortwave radiation](#) reaches the surface.

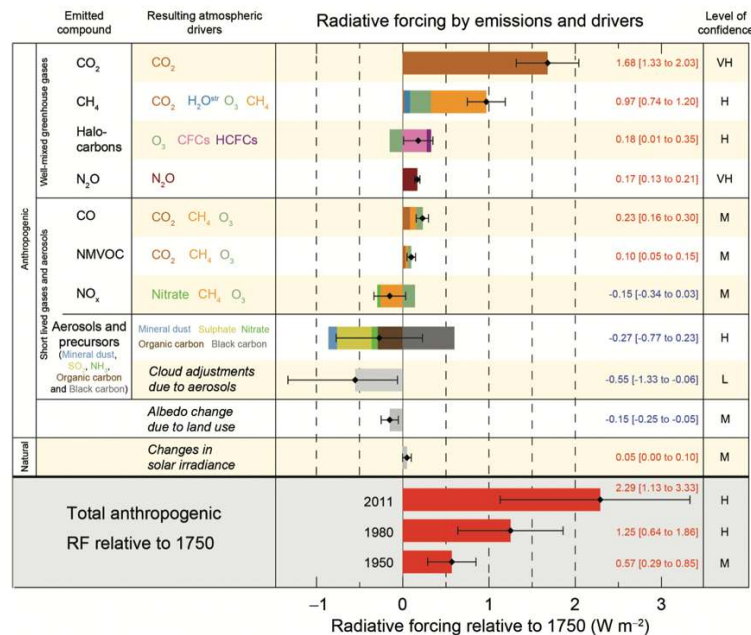
The exception is [black carbon](#) - its radiative forcing actually warms the Earth and also has a small impact when it is deposited on snow and ice (figure below left), decreasing the Earth's [albedo](#) slightly.



(Left) Ash ([black carbon](#)) from New Zealand's Mount Ruapehu covers the snow. This increases snow melt and decreases the surface [albedo](#). (Right) Beijing, October 9, 2010 covered by smog. (NASA image by Jeff Schmaltz, MODIS [Rapid Response](#).)

It should be noted that [feedbacks](#) from aerosols are still not understood as well as they need to be and their net impact on the Earth's radiation budget contains considerable [uncertainty](#) (see figure below, noting the large error bars on the aerosol values).

As a separate issue, the emission of industrial aerosols also has a severe impact on human health.



This figure shows the different things that have changed the radiative balance of the Earth since 1750. If the number is positive, it has the effect of warming the planet, if the number is negative the opposite is true. The horizontal black lines on each forcing show the range of [uncertainty](#) for each component. (Source: [IPCC AR5](#), [SPM](#))

Greenhouse gas emissions

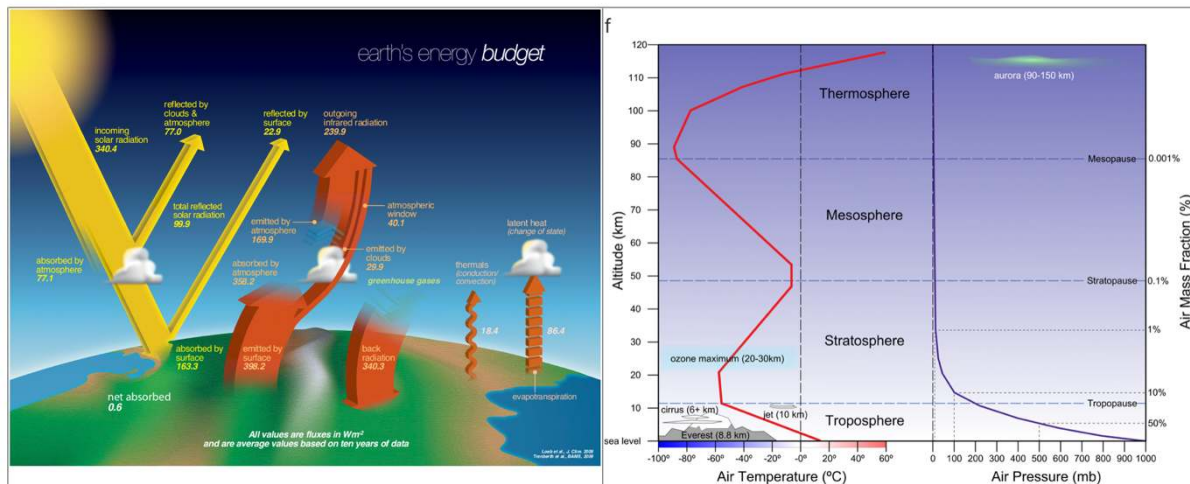
While the impact of aerosol emissions on climate is somewhat complex, the impact of human emissions of greenhouse gases is relatively simple: it increases the strength of the greenhouse effect.

To see this, recall from the [greenhouse effect lesson](#) that most of the Earth's emission of [longwave radiation](#) to space (that offsets the energy absorbed from the sun) comes from the atmosphere, rather than the surface (see figure below left, from that lesson). That is, the [longwave radiation](#) emitted by the Earth's surface (after absorbing sunlight and heating up) is absorbed by greenhouse gases in the atmosphere and reradiated (both back towards the surface, keeping it warmer than it would otherwise be, and out to space).

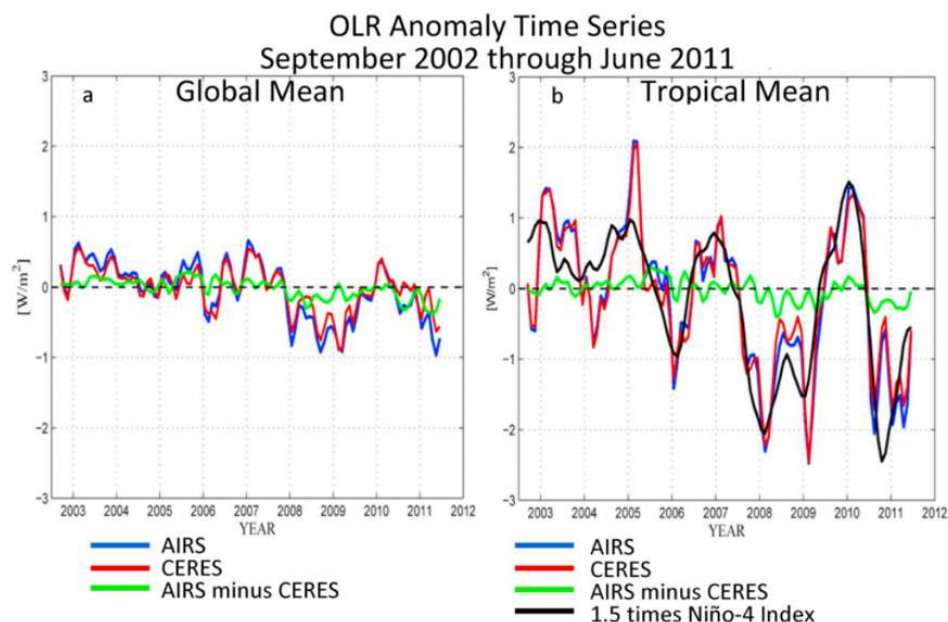
Also recall from the [atmospheric structure lesson](#) that most of the atmosphere is in the [troposphere](#) (about 80% of the mass / gas molecules are in the [troposphere](#), simply due to gravity), as can be seen in the figure below right from that lesson. The bulk of the Earth's emission of [longwave radiation](#) directly to space therefore comes from greenhouse gases somewhere in the middle of the [troposphere](#).

When we emit more greenhouse gases into the atmosphere, we increase the proportion of the atmosphere that is made up by greenhouse gases, making the atmosphere more opaque in the frequencies of [EM radiation](#) that greenhouse gases absorb (you might imagine this like more and more smoke in the air meaning you can't see as far).

This means that on average, the greenhouse gases that emit radiation to space are a little bit higher in the [troposphere](#) than they were before. From the figure below right, you can see that this means that the greenhouse gases emitting radiation to space are then slightly colder than before (since temperature decreases with height in the [troposphere](#) - the lowest section of the red line in the figure below right). As you learnt in the [EM radiation lesson](#), the colder a substance is, the less radiation it emits. So our addition of more greenhouse gases to the atmosphere means that slightly less radiation is being emitted to space by the Earth.



Since the amount of sunlight being absorbed by the Earth system is the same, but less radiation is being emitted to space, the Earth warms up. This is the [anthropogenic](#) greenhouse effect. Measurements of decreasing outgoing [longwave radiation](#) (OLR) from satellite, taken from a peer reviewed paper by Susskind et al (2012), are shown below.



The history of atmospheric CO₂ variability

Looking back over the last 400,000 years, [carbon dioxide](#) concentrations in the atmosphere have oscillated between about 180-300 [ppm](#) ([parts per million](#)), and are strongly correlated

with temperature (see Figure 1 below). The repeating cycles in this figure are [glacial-interglacial](#) cycles. The Last [Glacial](#) Maximum (ice age) occurred ~24,000-19,000 years ago. The transition from a [glacial](#) (colder) to [interglacial](#) (warmer) period, a deglaciation, is triggered by changes in [insolation](#) (the [Milankovitch cycles](#) we covered in the [Energy to Earth lesson](#)), and amplified by [feedbacks](#) within the Earth's [carbon cycle](#) (something we'll look at more in the [carbon cycle lesson](#) shortly). The last deglaciation ended about 10,000 years ago (we'll talk more about this in the paleo-climate lesson later in the course). Between then and the industrial revolution, we were in an [interglacial](#) state (the [Holocene](#)).



Figure 1 Atmospheric [CO₂](#) and surface air temperature variation over the last ~400,000 years (data from Vostock [Ice core](#), Petit et al., 1999)

As you can see from Figure 2 below, [anthropogenic](#) emissions of greenhouse gases including [CO₂](#) since the start of the industrial revolution are increasing atmospheric concentrations dramatically. We'll explore how we know that these increases are due to human activity shortly.

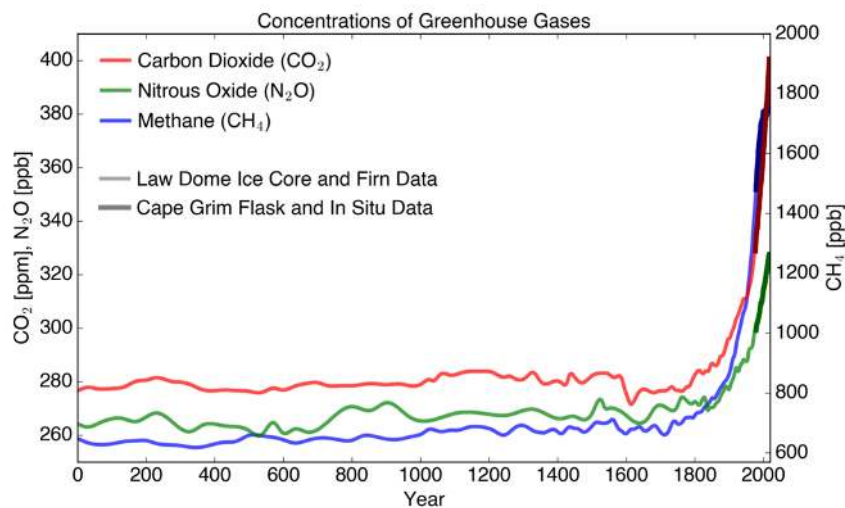


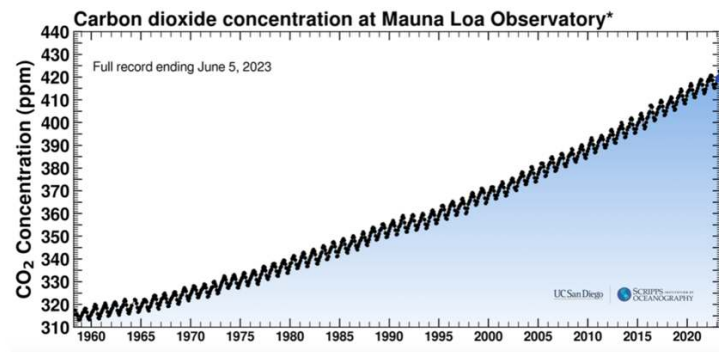
Figure 2 Common greenhouse gases in the atmosphere over the last 2000 years. Note that CH₄ and N₂O are in parts per billion.

The Keeling Curve

The Keeling Curve is a well known graph in [climate science](#) that shows direct measurements of atmospheric [CO₂](#) concentrations over time. It was named after Charles Keeling, an American scientist who took the first continuous measurements of atmospheric [CO₂](#) starting in 1958 at the Mauna Loa observatory in Hawaii. Mauna Loa was chosen as a good site to

measure [CO₂](#) because it is located at a high altitude and in the middle of a large ocean (the Pacific Ocean) so that the atmosphere is well mixed - the measurements give a good indication of [CO₂](#) concentration for the whole atmosphere.

Latest CO₂ reading: **424.58 ppm**

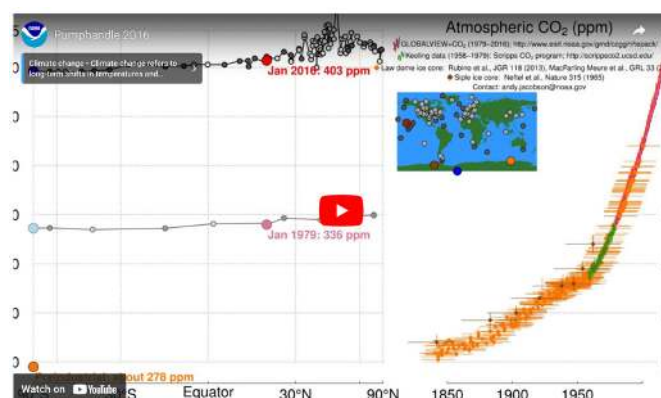


The Keeling Curve. Image courtesy of the Scripps Institute of Oceanography (click [here](#) for hourly updates of atmospheric [CO₂](#) levels and plots spanning anywhere from the last week to the last 800,000 years)

Looking at the figure above you should notice two things:

1. There is a clear upward trend over the last ~60 years and
2. There is some very predictable sub-annual [variability](#) in that trend. This [variability](#) is due to the seasonal uptake of [CO₂](#) by northern hemisphere vegetation. Have a look at the interactive video map showing "[A Year in the Life of the World's CO₂](#)". The video map will open in a pop-up window. Focus on the differences in northern hemisphere winter (Dec-Feb) vs northern hemisphere summer (June-Aug) and how the changing of the seasons affects atmospheric [CO₂](#) concentrations.

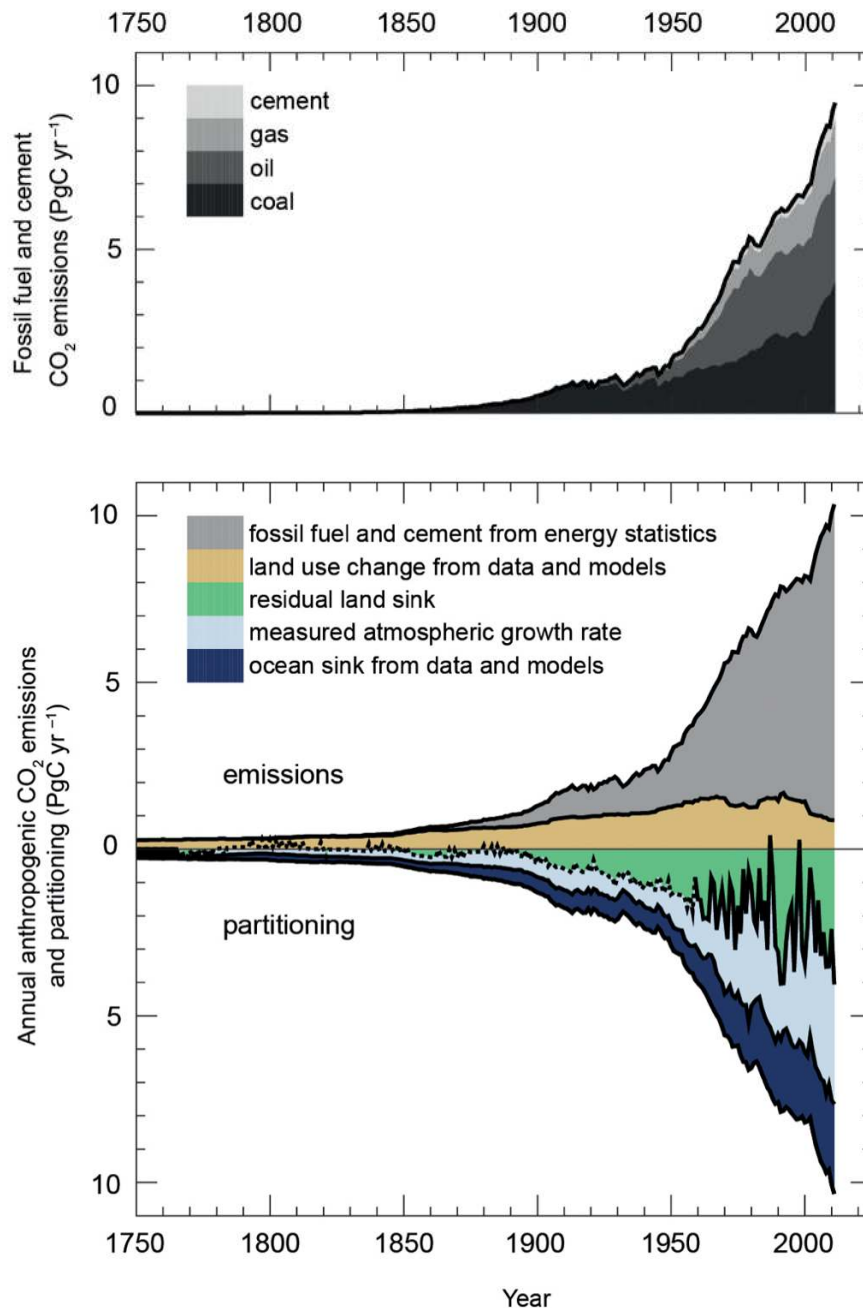
This upward trend is due to the [anthropogenic](#) release of [CO₂](#) primarily from burning fossil fuels. To see how this has changed over the 20th century watch the video below for a great summary of the last two pages of content. As you watch, look out for the seasonal cycle in [CO₂](#) - the way it goes up during austral summer and down in austral winter. Each dot on the graph is a different measurement station shown on the map to the right, with the Keeling Curve from Mauna Loa shown below it. See how measurements at Mauna Loa vary more than those at the south pole? Why do you think this is?



Evidence of human influence

How do we know that the increase in atmospheric [CO₂](#), as observed in the Keeling Curve, comes from [anthropogenic](#) emissions? There are several independent lines of evidence:

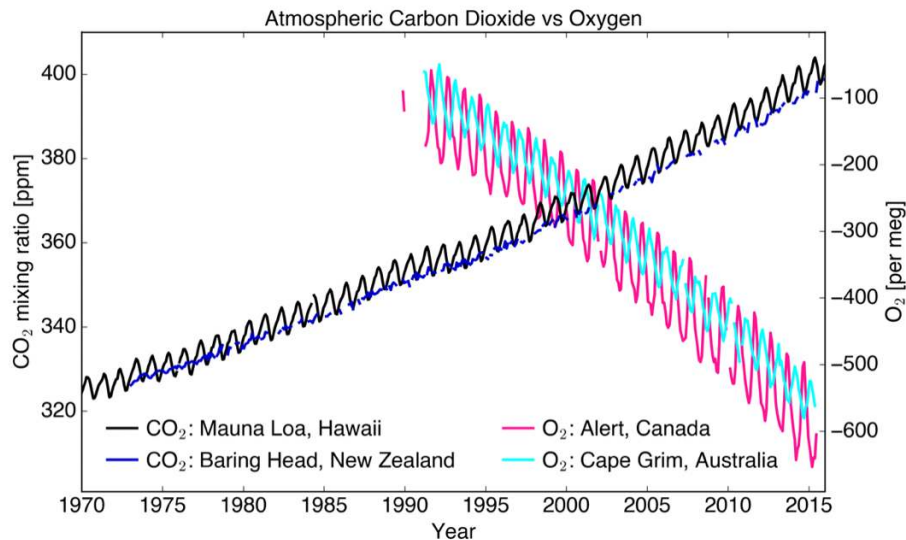
1. Account keeping - We have a reasonably good idea of how much carbon we've emitted since the industrial revolution (about 545 ± 55 [GtC](#) as of 2015) and this is more than enough to explain the increase we've seen in atmospheric [CO₂](#).



[IPCC, AR5](#)

2. Changes in atmospheric oxygen. When we measure oxygen in the atmosphere, we can see that it has decreased in a way that is consistent with the burning of carbon based fuels

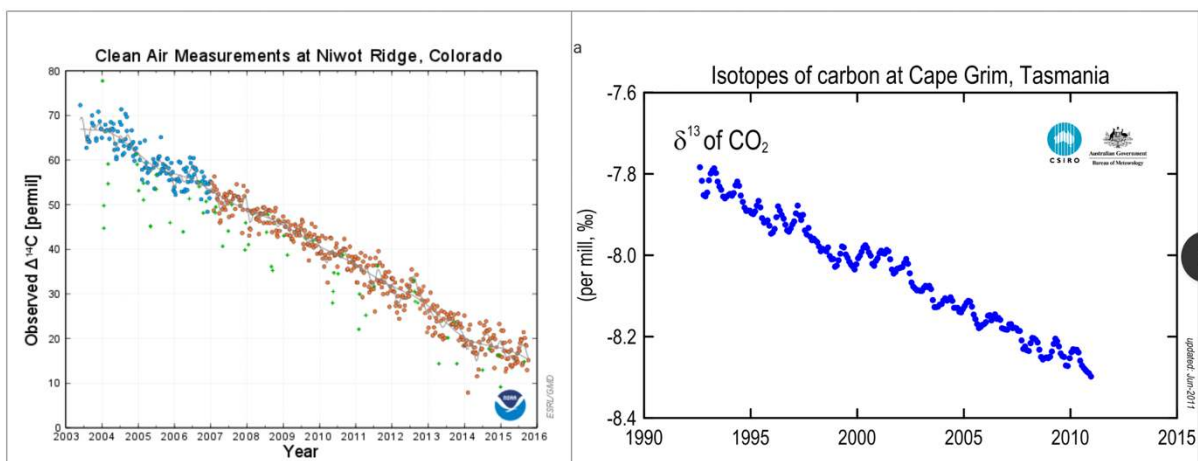
(see figure below). C and O₂ join in the combustion to make [CO₂](#) - the process consumes oxygen.



Changes in atmospheric [CO₂](#) and O₂ over the last 45 years (note, oxygen is measured in per meg, where 1 per meg is equal to 0.0001 [ppm](#))

3. Isotopic composition of atmospheric carbon 14. Most carbon [atoms](#) have a molecular weight of 12, but a small percentage have weight 13 or 14. Carbon 14 is a radioactive carbon [isotope](#) that has a [half life](#) of about 6000 years, and it is used to date the age of [organic carbon](#)-containing material (there's more about what an [isotope](#) actually is in [additional material this week](#)). Generally speaking, the less carbon 14 something has, the older it is. Fossil fuels are the result of plant matter breaking down and transforming into coal and oil over millions of years, and since these plants took their [CO₂](#) from the atmosphere (where carbon 14 is created) a long, long time ago, the resulting oil and coal are very low in carbon 14 - it having decayed away over this time period.

If we look at the prevalence of carbon 14 in [CO₂](#) molecules in the atmosphere over time, we can see that it is decreasing as [CO₂](#) increases (see left hand side figure below). So the majority of whatever is being burnt is high in carbon 12 and low in carbon 14 and was therefore alive a long, long time ago (e.g. it's not a result of today's forests burning).



4. Isotopic composition of atmospheric carbon 13. The proportion of carbon 13 is also declining (figure above right), suggesting that what is being burnt is high in carbon 12 and low in carbon 13. Since the [photosynthetic](#) pathway in plants preferences lighter carbon [isotopes](#) (C^{12}), plants have a low proportion of carbon 13 compared to other sources, suggesting that what is being burnt is plant-based material.

So to summarise, the source is likely something plant-based, that is very very old, and is being burnt to produce the [CO₂](#), and we know that we have burnt enough fossil fuels to easily account for the atmospheric increase in [CO₂](#).

GW Potential

[Carbon dioxide](#) is the greenhouse gas with the greatest [anthropogenic](#) impact on the climate, however it is not the only one we emit. The different gases we put in the atmosphere have greatly varying impacts on the Earth's radiation budget and this is due to three main reasons.

1. How efficient that gas is at capturing [longwave EM radiation](#);
2. The amount of the gas we are emitting into the atmosphere relative to its natural concentration and finally,
3. How long a gas remains in the atmosphere in its original molecular state - for example, [methane](#) is a very potent greenhouse gas but most of it will break down into other chemicals ([CO₂](#) and [water vapour](#)) after several years. By breaking down or being taken up by a different part of the [climate system](#), these greenhouse gases will have a different impact on the Earth's radiation budget on different time scales.

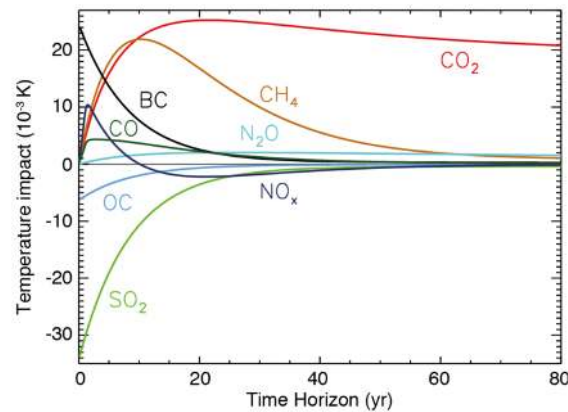
We try to combine the differences in the efficiency, concentration and atmospheric lifetime of different greenhouse gases into a common measure, called [Global Warming Potential \(GWP\)](#), that allows us to more easily compare the effect of emissions of different greenhouse gases. [GWP](#) compares the overall effect of emitting different greenhouse gases relative to the effect of emitting [CO₂](#). [CO₂](#) by definition has a [GWP](#) of 1 while [methane](#), for example, has a [GWP](#) of 28 - emitting one tonne of [methane](#) will have about 28 times the effect of emitting a tonne of [CO₂](#) (over a period of 100 years). By doing this for each greenhouse gas, we can compare them on the same scale of impact.

Below is a table of the most common greenhouse gases ([CO₂](#), CH₄ and N₂O) that we emit, as well as two rarer ones (CFC-13 and PFC-14). It should be noted that these numbers are not exact - finding the [GWP](#) of these gases in the real world is very difficult so estimates vary.

Common name	Chemical Formula	Atm. concentration	Lifetime (Years)	GWP (20 Years)	GWP (100 Years)
Carbon dioxide	CO ₂	400 ppm	3-5*	1	1
Methane	CH ₄	1834 ppb	12.4	84	28
Nitrous Oxide	N ₂ O	328 ppb	121	264	265
Chlorotrifluoromethane (CFC-13)	CClF ₃	2.7 ppt	640	10900	13900
Perfluoromethane (PFC-14)	CF ₄	80.7 ppt	50000	4880	6630

[Anthropogenic](#) greenhouse gases and their [Global Warming Potential \(GWP\)](#) over both 20 and 100 years. Note that while [CO₂](#) is shown having an atmospheric lifetime of just a few years, the impact of the imbalance it creates in the [carbon cycle](#) means its impact is actually on the order of centuries rather than just a few years. This is also illustrated in the figure below.

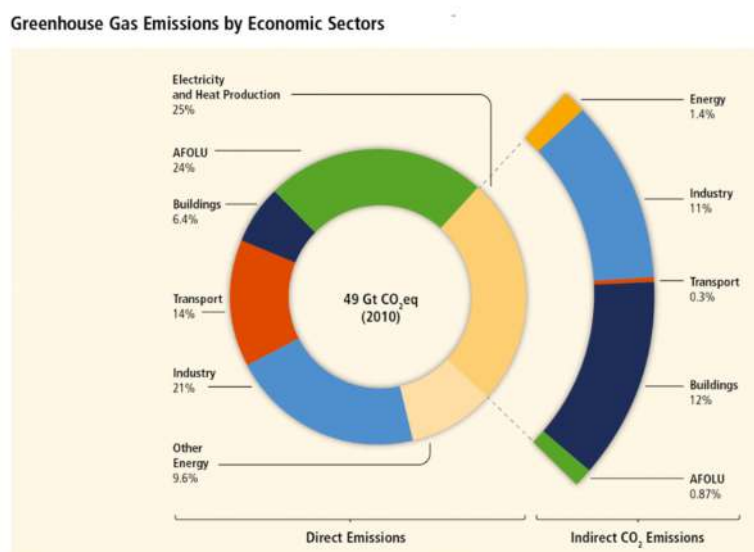
So, while other gases we emit are far more efficient at trapping [longwave radiation](#), our concern is primarily regarding [CO₂](#) and this is because it is the gas we emit in the greatest quantities. In addition, its impact on the radiative balance of the planet can last for centuries, due to the imbalance our emissions create in the [carbon cycle](#) (which we'll look at shortly). To visualise this, see the figure below.



This figure shows the impact of different [anthropogenic](#) gases/ emissions on Earth's temperature over time if they were emitted at a single point in time (time = 0) and over one year only, and gives us an indication of how long each gas will have an effect on temperature. Taken from [IPCC AR5](#). [CO₂](#) = [carbon dioxide](#); [CH₄](#) = [methane](#); [N₂O](#) = [nitrous oxide](#); [NO_x](#) = [nitrogen oxide](#); [SO₂](#) = [sulfur dioxide](#); [OC](#) = [organic carbon](#); [BC](#) = [black carbon](#); [CO](#) = carbon monoxide.

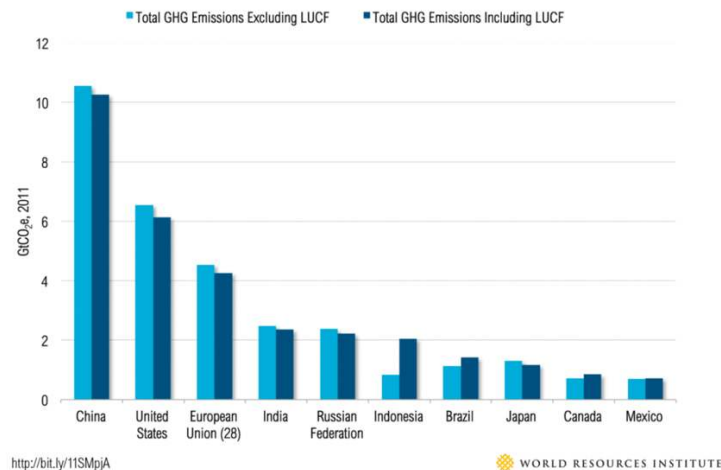
Emissions by sector

So, we know that [CO₂](#) is a potent greenhouse gas and that we are emitting more of it than ever before in human history, but where is it coming from and who is doing it? The first answer is simple - most of the [CO₂](#) we emit comes from the generation of electricity (25%), followed closely by agriculture and land use (24%) and industrial emissions (21%) which make up the bulk of emissions. The domination of electricity generation as a source of both direct and [indirect emissions](#) is one reason why transitioning to renewable energy sources is considered to be the most important path to a zero carbon future.



Direct and [indirect emissions](#) by sector in 2010. AFOLU stands for Agriculture, Forestry and Other Land Use. Units of Gt [CO₂](#) equivalent (standardising all the [GHG](#) emissions to the common [GWP](#) of [CO₂](#) [Figure SPM.2 from the Summary for Policymakers in Working Group 3 of the 5th [IPCC](#) Assessment Report])

As for countries, perhaps unsurprisingly, the world's two biggest economies are also the two biggest emitters, with China overtaking the USA within the last decade. China is, however, also leading the world on the implementation and deployment of renewable energy, in part to offset the problem they face with pollution (see [aerosols page](#)).

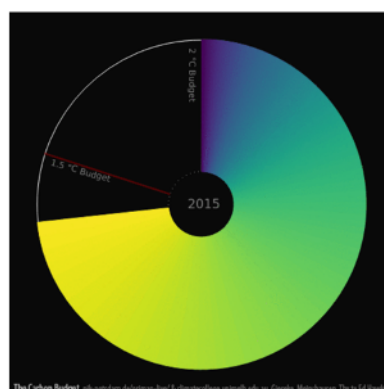


Finally [this link shows an interactive summary of everything presented here](#): sources of greenhouse gases and who emits what as well as emissions per capita (something Australia is quite high in the rankings for).

The carbon budget

Moving into the future, our greatest concern is limiting the amount of warming to 2 degrees Celcius (ideally 1.5 degrees, according to the [Paris agreement](#)). From 1870 to 2011, it is estimated that humanity has emitted 400 [gigatonnes](#) of carbon into the atmosphere.

If we want a good chance of avoiding warming of 1.5 or 2 degrees, there is only so much more [CO₂](#) we can emit into the atmosphere before warming is guaranteed to go above those temperatures. The figure below animates that 'carbon budget', showing the cumulative emissions of [CO₂](#) over time and how we are already approaching the point where 1.5 degrees of warming is certain.



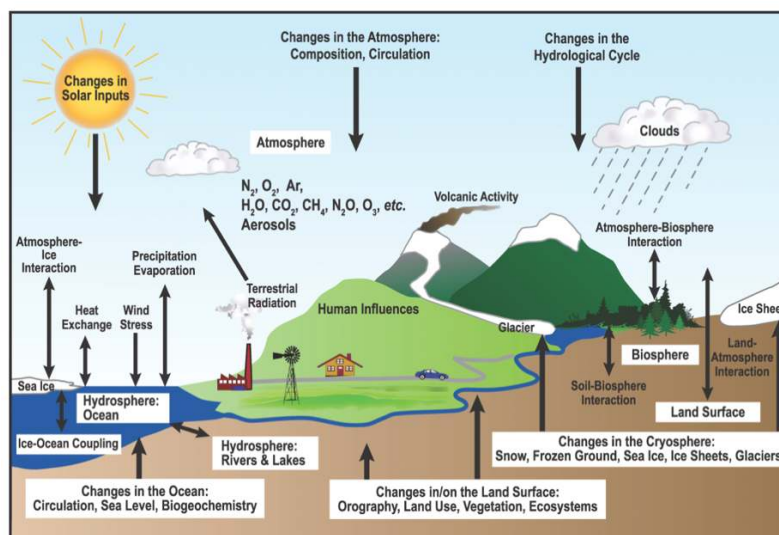
2 - Interactions in the climate system

Introduction

A stable climate over the last 10,000 years has almost certainly helped the development of human civilisations. But what do we mean when we say a stable climate, and how does the climate change from being stable to unstable? The answer to these questions lies in the interactions between different parts of the [climate system](#) and how these interactions either amplify or suppress the effect of a perturbation to the [climate system](#). This lesson will present an overview of this very complex topic and, as usual, provide you with extra material if you wish to look into it in more depth.

What is climate?

The [climate system](#) is a combination of several 'spheres' including the atmosphere, [hydrosphere](#) (liquid water), [cryosphere](#) (ice and snow), [geosphere](#) (the rocks and land surface of the Earth) and the [biosphere](#) (living things). These spheres interact with each other by exchanging energy (e.g. a warm ocean heats the atmosphere above it), [momentum](#) (think of strong winds shifting sand dunes or making waves), mass (e.g. melting [glaciers](#) transfer mass from the [cryosphere](#) to the [hydrosphere](#)), carbon, oxygen and much more. These processes - there are too many to list here - are ongoing and driven primarily by the energy we receive from the sun.



A simple schematic showing the components of the [climate system](#) and some examples of how spheres interact with each other

So what is climate? It can be thought of quite simply as the average of weather over a period of many years (typically about 30 years). Watch the two videos below to get the basic idea, and if you want to know more, check out this link from [NASA](#).

The purpose of showing you this is to emphasise that some interactions between these spheres within the [climate system](#) vary on short time-scales (hours, days weeks, even years) but are effectively unchanging on longer time scales if we consider their average over multiple decades. Defining climate as the average state over many years removes the [noise](#) of weather and other short term [variability](#) in the [climate system](#), and allows us to focus on drivers that cause changes to this longer term average state. These drivers might include the

changes in incoming solar radiation we discussed in previous lessons, or changes in greenhouse gas concentration. We are particularly interested in how a change in these drivers - effectively a perturbation to the [climate system](#) - change the interactions between these spheres.

In summary, 'climate' is nothing more than a statistical description of weather over a longer period of time, typically a few decades. If you'd like to know more, Skeptical Science has written up a more [detailed primer](#) on the components of the [climate system](#) and why each one is important.

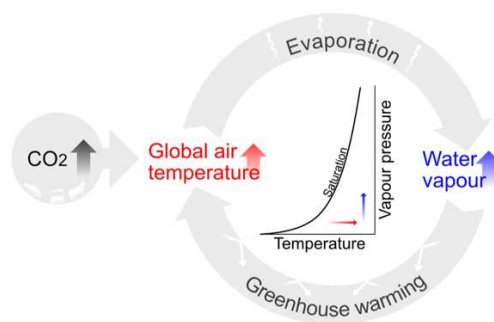
Climate Feedbacks

The interactions between the atmosphere, [biosphere](#), [hydrosphere](#), [cryosphere](#) and [geosphere](#) that define the [climate system](#) might change if there is a change to the 'external forcing' of the [climate system](#). A change to the external forcing might, for example, be a change in the amount of sunlight reaching the Earth because of change in solar activity, the eruption of a very large volcano that releases aerosols into the [stratosphere](#) (cooling the climate), or large scale emission of greenhouse gases from fossil fuel burning. Each of these represents a perturbation to the [climate system](#) by changing the amount of energy in the system.

A [feedback](#) is a collection of processes that either amplify or suppress the effect of a perturbation to the [climate system](#). When the system amplifies the effect of an initial perturbation, it's called a positive [feedback](#), when the system suppresses the effect of an initial perturbation, it's called a negative [feedback](#). Below we have schematics that demonstrate a positive and negative [feedback](#) in the [climate system](#), both of which are currently occurring due to [anthropogenic](#) warming. Below, we have the [water vapour feedback](#) and it works as follows:

1. [CO₂](#) increase in the atmosphere raises air temperatures
2. This increase in air temperature leads to more evaporation
3. This leads to more [water vapour](#) in the atmosphere (remember, [water vapour](#) is an efficient greenhouse gas)
4. The increased [water vapour](#) causes more greenhouse warming, this now loops back to step two and the cycle continues.

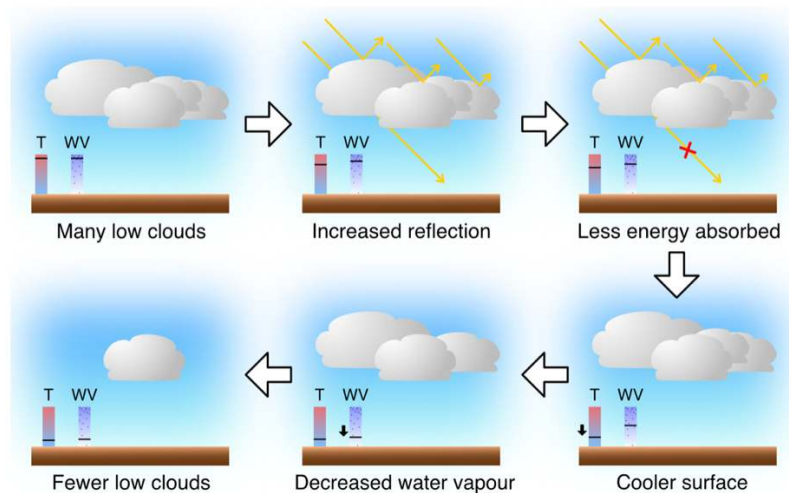
This is a positive [feedback](#) as the initial perturbation - an increase in air temperature due to [CO₂](#) - is re-enforced by the impact of this change: warming leads to more [water vapour](#) in the air which leads to more warming.



The [water vapour feedback](#)

Next let's look at a negative [feedback](#), the cloud [albedo feedback](#) (below).

1. Increased warming means more [water vapour](#) in the atmosphere - this means an increase in low-level clouds.
2. Because clouds are white, they have a high [albedo](#). This extra cloud cover now reflects more sunlight back into space.
3. As less [shortwave](#) energy reaches the surface, it cools.
4. A cooler surface means less evaporation and therefore, less [water vapour](#) in the atmosphere.
5. Less [water vapour](#) in the atmosphere means fewer low clouds.



The cloud [albedo feedback](#)

This is a negative [feedback](#) because the mechanism at work here (increased low-level cloud cover) starts a process that opposes the effect of the initial perturbation. To put it simply, the increase in cloud cover due to warming leads to surface cooling which reduces cloud cover.

These are but two of many interconnected [feedbacks](#) in the [climate system](#) (we showed you a figure illustrating the ice-[albedo feedback](#) in the introduction, too) and the way they interact with each other over time determines climate stability and its sensitivity to change.

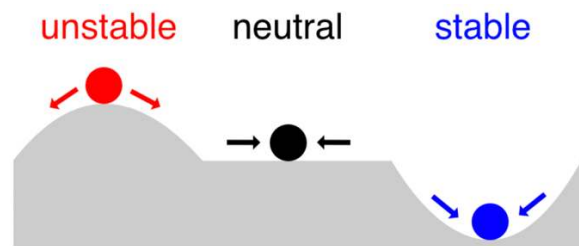
Climate stability

So we've seen that the [climate system](#) is made up of many interactions between its different components, and that these can act to amplify or suppress perturbations to the [climate system](#). When we talk about [climate system](#) 'stability', we are essentially interested in whether [climate system](#) responses to perturbations are dominated by negative [feedbacks](#) (in which case the system could be said to be stable) or positive [feedbacks](#) (in which case the system could be said to be unstable), or something in between.

In the figure below, the three balls represent different climate states: unstable, neutral and stable. In the stable state, if we perturb the system away from its equilibrium state (up to the sides of the bowl it's sitting in), it simply rolls back down to where it started. The stable state is just that: one that when impacted by an external force seeks to return to its prior equilibrium. It is for this reason that stable states are dominated by negative [feedbacks](#) - remember, these are the [feedbacks](#) that oppose any changing force in the system.

The neutral state represents a climate where external forcing has an effect on the average climate, somehow in proportion to the size of the forcing. Another way to think about this is if we doubled the size of the forcing change to the system, we'd see twice the size of the response.

In the unstable state, even a small perturbation might cause the ball to roll down the hill, away from its previous equilibrium. An unstable climate response would therefore be dominated by positive [feedbacks](#). Remember, once a positive [feedback](#) begins, it amplifies itself much in the same way that pushing a ball down a hill means that the ball will simply keep rolling on its own after you give it the initial push.



These three balls represent different climate states: unstable, neutral and stable.

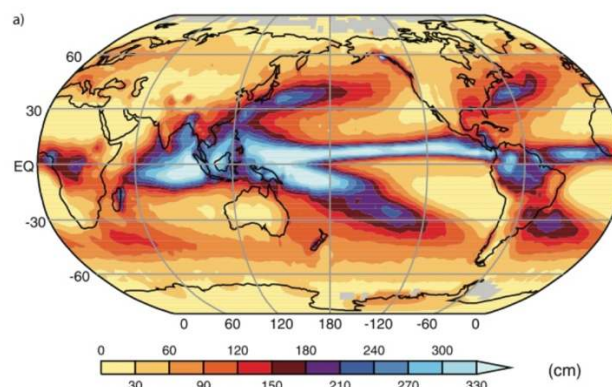
For the last part of this section, watch the video below. It gives excellent examples of how the climate of the last few millions of years can be thought of as bistable between ice age or not ice age. It also ties in the ideas of [feedbacks](#) in a visually easy to follow way.

3 - Water in the atmosphere

Introduction

In the last two lessons we looked at how [anthropogenic](#) activity is perturbing the [climate system](#), and how the [climate system](#)'s response to perturbations is complicated by [feedbacks](#) within the [climate system](#). In this lesson and the next, we'll look very briefly at some processes that play a key role in major [climate feedbacks](#).

In this lesson, we'll do this by trying to understand why there is so much rainfall around the Earth's equatorial regions (see figure below, showing average yearly precipitation, from the [IPCC](#)).



Remember from the [Energy to Earth lesson](#) that the Earth absorbs more of the sun's energy around the equator, so we know that there is more energy available in the [tropics](#) - it's hotter there. This is only part of the story...

Water Phases

So we all understand that water can exist as a solid, liquid or gas. What you might not immediately realise is that the gas - [water vapour](#) - is totally transparent, so that steam, clouds and fog are actually droplets of water in the air, rather than gas. In the image below you can see hot [water vapour](#) (clear) condensing into steam as it cools.



So, can [water vapour](#) exist below 100°C? Yes. [Water vapour](#) in the air is what we call humidity, and it is measured in many different ways.

Humidity

Humidity describes how much [water vapour](#) is in the air. It measures in several different ways.

Specific humidity is a measure of how much [water vapour](#) is in the air (in grams, or kilograms) relative to the total amount of air (in grams, or kilograms), so it has units of grams per gram or kg per kg. **Vapour pressure** (more accurately 'partial vapour pressure') measures how much pressure [water vapour](#) contributes to the total air pressure.

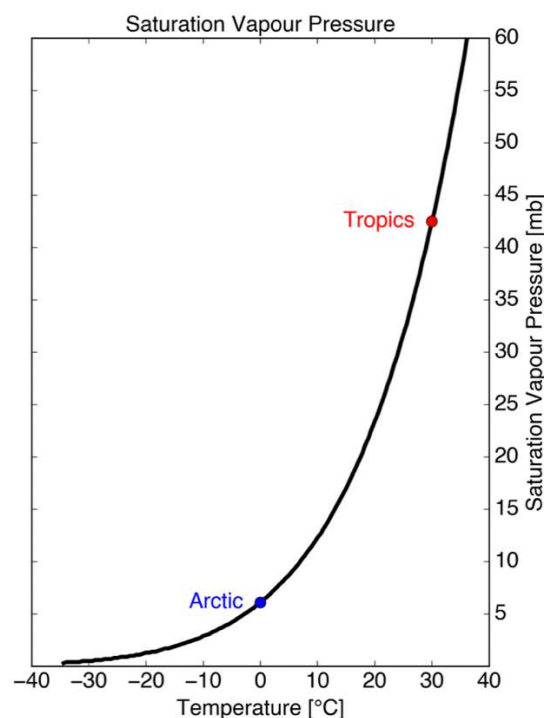
There are however some physical limits on the amount of [water vapour](#) that can be in the air. If [water vapour](#) concentrations become too large, many more of the water molecules interact with each other and start to change to the liquid phase - water droplets form. For a given temperature of the air, there is a threshold amount of [water vapour](#) in the air where this happens, called the **saturation vapour pressure**. When more [water vapour](#) is added (increasing the vapour pressure above this threshold) water in the air will start to condense into water droplets. This threshold value is strongly dependent on the temperature of the air (see figure below).

So the maximum amount of [water vapour](#) that could potentially be in a [parcel](#) of air depends solely on the temperature of the air, following the relationship in the graph below. This is often explained as the 'amount of [water vapour](#) air can hold' being dependent on temperature, and while this is a very simple (and probably useful) way to think about what is going on here, [it's not quite right](#). I will not examine you on this distinction!

Relative humidity is a measure of how much [water vapour](#) is in a [parcel](#) of air, relative to how much [water vapour](#) could potentially be in that [parcel](#) of air, given its temperature. For example, if a [parcel](#) of air has a relative humidity of 60%, it means that the air at this temperature could still have another $40/60 = 67\%$ more [water vapour](#) before it would start to form dew, clouds, or fog (there is a more detailed explanation on the next two pages).

If we now cool that [parcel](#) of air but keep the amount of [water vapour](#) constant, its relative humidity would increase as cooler air has a lower maximum [water vapour](#) threshold. This could result in the same [parcel](#) of air going from 60% relative humidity to 90% relative humidity, while still having exactly the same amount of [water vapour](#) in it (i.e. the same specific humidity). If we cool it further, it would reach 100% relative humidity and water droplets would start to form, as might happen at the end of a still clear day when dew starts to form.

Note that relative humidity can also be expressed as the ratio of partial vapour pressure to saturation vapour pressure.



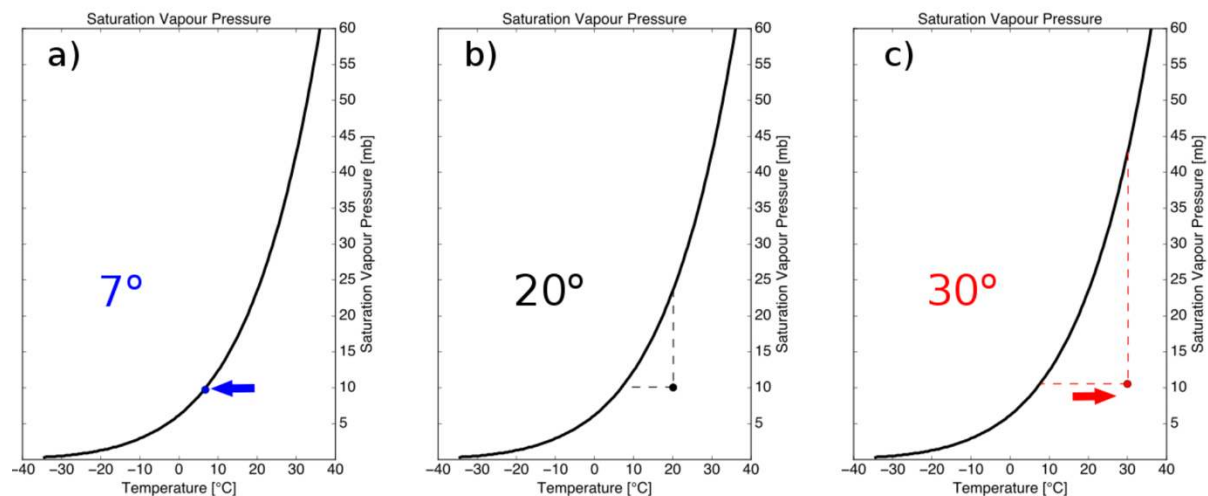
A figure showing temperature vs saturation vapour pressure. The black line signifies the maximum amount of [water vapour](#) that a [parcel](#) of air can hold at a given temperature. Above that line, water droplets will form as the [parcel](#) is now 100% saturation. Below the line, the [parcel](#) is less than 100% saturated. Note again how at cooler temperatures, the air can hold less water (ie: lower saturation vapour pressure).

Humidity example

Let's say we have a [parcel](#) of air at a temperature of 20°C that has a vapour pressure of 10 [mb](#) (shown in Figure b below). This [parcel](#) of air is about 43% saturated (it has a relative humidity of 43%) at this temperature, since its saturation vapour pressure is ~23 [mb](#). Suppose we now decrease the temperature of that [air parcel](#) without changing how much [water](#)

[vapour](#) it contains (Figure a). If we reduce the temperature to 7°C, the [parcel](#) is now completely saturated at 10 [mb](#) vapour pressure and therefore, the air is at 100% relative humidity.

What if instead, we increased the temperature of the [air parcel](#)? If we heat it to 30°C, the air is able to hold far more [water vapour](#), with a saturation vapour pressure of about 42 [mb](#) (Figure c). This means that the amount of [water vapour](#) in the [parcel](#) of air is only 25% of what it could hold at that temperature (remember, we aren't changing the amount of [water vapour](#), 10 [mb](#), only the temperature) - meaning that the relative humidity is now only 25%. So even if the amount of [water vapour](#) in the air doesn't change over the course of a day, changes in the temperature of the air mean that the relative humidity can vary a great deal.



In summary, the temperature of an [air parcel](#) determines the maximum amount of [water vapour](#) that can be in the air. Without changing the [water vapour](#), if we increase the temperature, we decrease the relative humidity. Similarly, if we decrease the temperature, we increase the relative humidity.

The ideal gas law

For a given [parcel](#) of air, there are strict relationships between its temperature, pressure and volume. These are expressed in what's called the *Ideal Gas Law*. While we won't ask you to use the equation, we do need you to understand what it's about. The essence of the relationship is that Temperature is proportional to Pressure times Volume (think of $T \sim P \times V$) - if the temperature of a [parcel](#) of air increases, either its pressure or volume (or both) must increase.

What this means in practice is that when the sun heats one part of the Earth's surface more than another (e.g. a darker / lower [albedo](#) region):

- that area heats the [parcel](#) of air above it more than in other areas
- when this [parcel](#) of air becomes warmer, the volume it occupies increases
- this [parcel](#) of air is now less dense than surrounding air and, like an air bubble in water, will rise upward through the atmosphere
- as the [air parcel](#) rises, it begins to cool because its pressure drops

Convective cloud formation

So, tying all of this together (and repeating some of what you just read):

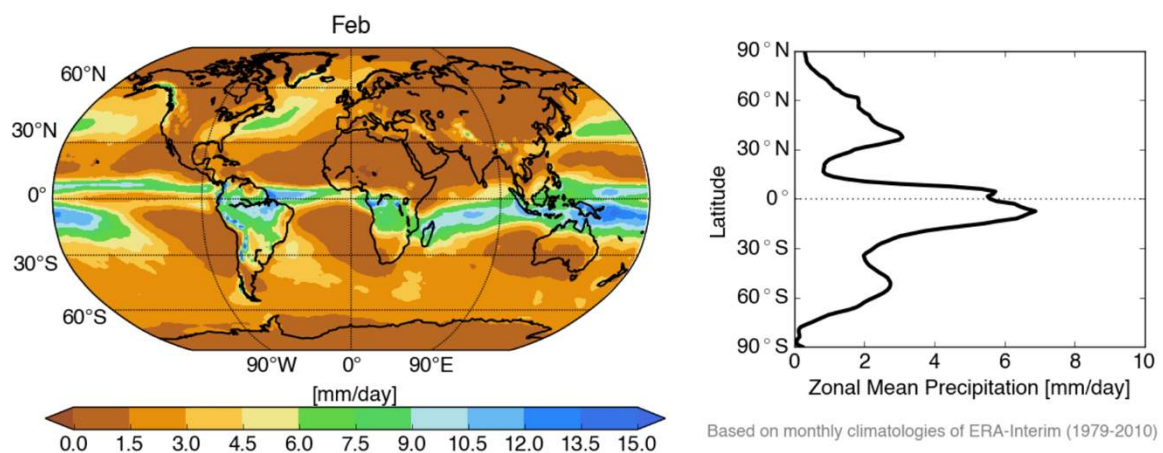
1. the sun heats some parts of the Earth's surface more than others
2. for those areas, the [parcel](#) of air above them heats up more than in other areas
3. when this [parcel](#) of air becomes warmer, the volume it occupies increases
4. this [parcel](#) of air is now less dense than surrounding air and, like an air bubble in water, will rise upward through the atmosphere
5. as the [air parcel](#) rises, it begins to cool because its pressure drops
6. as it cools, the relative humidity in that [parcel](#) of air can reach 100%, forming cloud droplets
7. if the air in a region has similar humidity, this results in a collection of puffy clouds with flat bottoms

The inter-tropical convergence zone (ITCZ)

We're now in a position to answer the question asked at the start of this lesson: why is there so much more rain at the equator than elsewhere?

1. We know it's hotter there, so the air has a much higher amount of [water vapour](#) in it.
2. The sunlight is stronger there than elsewhere, driving [convection](#) of the type we just discussed
3. [Convection](#) of very moist air means lots of rain

The animation below gives you an idea of the [climatology](#) of rainfall across the globe (i.e. what average rainfall looks like for each month of the year).



This large scale [convection](#) results in huge cells of air rising near the equator where it is warmest. These then move towards the poles high up in the atmosphere, cool, and descend back down to Earth in the mid latitudes. These are known as the Hadley cells - something we'll cover in a future lesson.

The rising air in the [tropics](#) sucks in vast amounts of air below it, pulling in more moist air and causing north-easterly and southeasterly [trade winds](#) in the northern and southern hemispheres (again, something we'll cover later). These winds converge in this area of tropical [convection](#) - the [Intertropical Convergence Zone](#), or [ITCZ](#). Much like with the

seasonality of surface temperatures, the location of the [ITCZ](#) also shifts with the seasons (see animation above).

4 - The Carbon Cycle

Introduction

The [Carbon cycle](#) is an integral part of the Earth's [climate system](#), and describes the exchange of carbon between different spheres of the [climate system](#). In this lesson you'll learn about the main sources (things that release [CO₂](#) into the atmosphere) and sinks (things that absorb [CO₂](#) from the atmosphere) in the [climate system](#), and how increasing [anthropogenic](#) carbon emissions from fossil fuels affects the [carbon cycle](#).

Carbon stored in the Earth system

In the figure below you can see estimates of the amount of carbon stored in different parts of the Earth system. The units for these numbers are [gigatons](#) of carbon (note that similar diagrams might have numbers that are about 3.7 times higher if they are expressed as [gigatons](#) of [CO₂](#) instead, since the ratio of the molecular weights of [CO₂](#)/C is about 44/12).

Note that the land surface, which includes carbon trapped in soil and plants, contains about 2900 Gt. Fossilised carbon, including that potentially extractable as fossil fuels, is about 10,000 Gt. The atmosphere, which contains carbon in the form of [CO₂](#), holds only around 800 Gt.

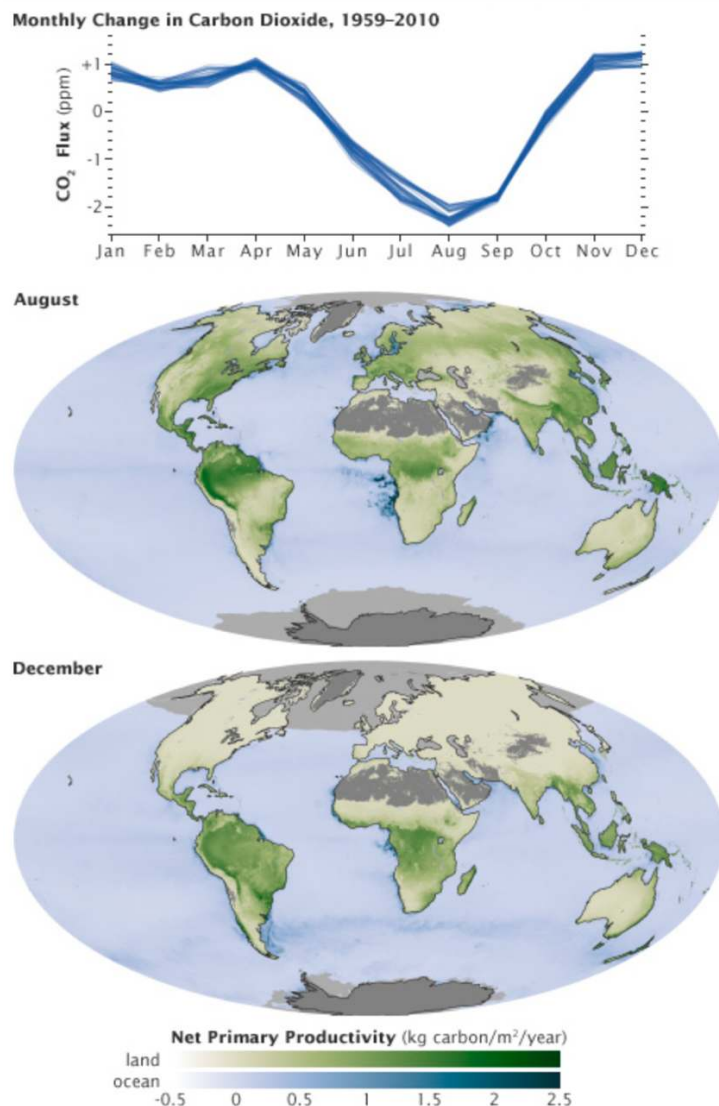
Finally, the ocean contains an order of magnitude more carbon than both the atmosphere and land surface combined, an estimated 38,000 Gt, in the form of both [dissolved inorganic carbon](#) and [organic carbon](#) from marine creatures.

The takeaway message? The land contains more than 3 times more carbon than the atmosphere and the ocean nearly 50 times more. If this carbon were somehow released into the atmosphere it would have considerable consequences for the strength of the greenhouse effect. Understanding the stability of these carbon stores, and the [carbon cycle](#) of which they're a part, is at least potentially very important.

Carbon exchange with the land surface

Carbon is constantly exchanged between the ocean and atmosphere, and land surface. Clicking on [this link](#) will take you to a NASA animation of the Earth's annual atmospheric [CO₂](#) cycle. As is described in the animation, the Earth usually cycles between a low and high [CO₂](#) value once a year - as you saw in the Keeling curve - driven by changes in [primary productivity](#) of the [biosphere](#), dominated by the northern hemisphere. As we can see in the figure below, atmospheric [CO₂](#) begins to decrease during the northern hemisphere spring as plants begin to grow again, taking up carbon from the atmosphere during the process of [photosynthesis](#). It reaches a minimum in northern hemisphere summer, which can be seen in the Figure for [primary productivity](#) in August. Note the green patches in North America, Europe and Asia. From August to December, this process reverses as trees and plants die back in northern hemisphere Autumn. The carbon from both decomposition of organic matter and plant respiration is released back into the atmosphere, so that we reach a maximum [CO₂](#) value around December. If at this point you're wondering why [CO₂](#) follows

northern hemisphere seasons so closely, it is simply because that's where most of the land on Earth is, and therefore most of the land-surface [biosphere](#).



Net [primary productivity](#) on land and in the ocean at different times of year (source - [NASA](#))

The land surface also contains [permafrost](#) - large areas where the ground is frozen below a certain depth in the soil, even in summer. Climate scientists are concerned that when this [permafrost](#) melts as the climate warms, the organic matter contained in these soils will decompose in the water that thaws, releasing large quantities of [methane](#) (a potent greenhouse gas). This would further warm the the climate - a positive [feedback](#).

Carbon exchange with the ocean

Carbon exchange with the ocean involves two main mechanisms: the 'solubility pump' and the 'biological pump'.

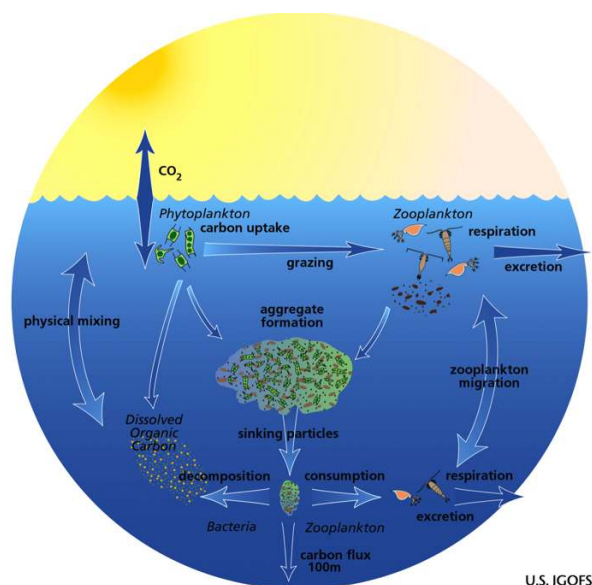
The solubility or 'physical' carbon pump refers to [CO₂](#) dissolving into the ocean or outgassing from it simply because there is a difference in the [CO₂](#) concentration between the air and water. Aside from this air-sea [gradient](#) of [CO₂](#), other factors such as wind speed (which

affects ocean mixing) and the temperature of the water change the rate at which this occurs. Once in the ocean, the CO_2 forms carbonic acid (H_2CO_3) which then further breaks down into a hydrogen ion (H^+) and bicarbonate (HCO_3^-). Some of the bicarbonate then breaks down into another hydrogen ion (H^+) and carbonate (CO_3^{2-}). The three forms of carbon (dissolved $\text{CO}_2(\text{aq})$, carbonate and bicarbonate) that result from these reactions are collectively known as [Dissolved Inorganic Carbon \(DIC\)](#).

To those of you familiar with chemistry, you'll recognise that this increase in H^+ (hydrogen ions) results in a lowering of the [pH](#) of the water: it becomes more acidic. While you'll learn about the impacts of ocean acidification in a later lesson, this is essentially its root cause; increasing atmospheric CO_2 is increasing the air-sea [gradient](#) of CO_2 , resulting in more uptake by the ocean, increasing ocean acidity.

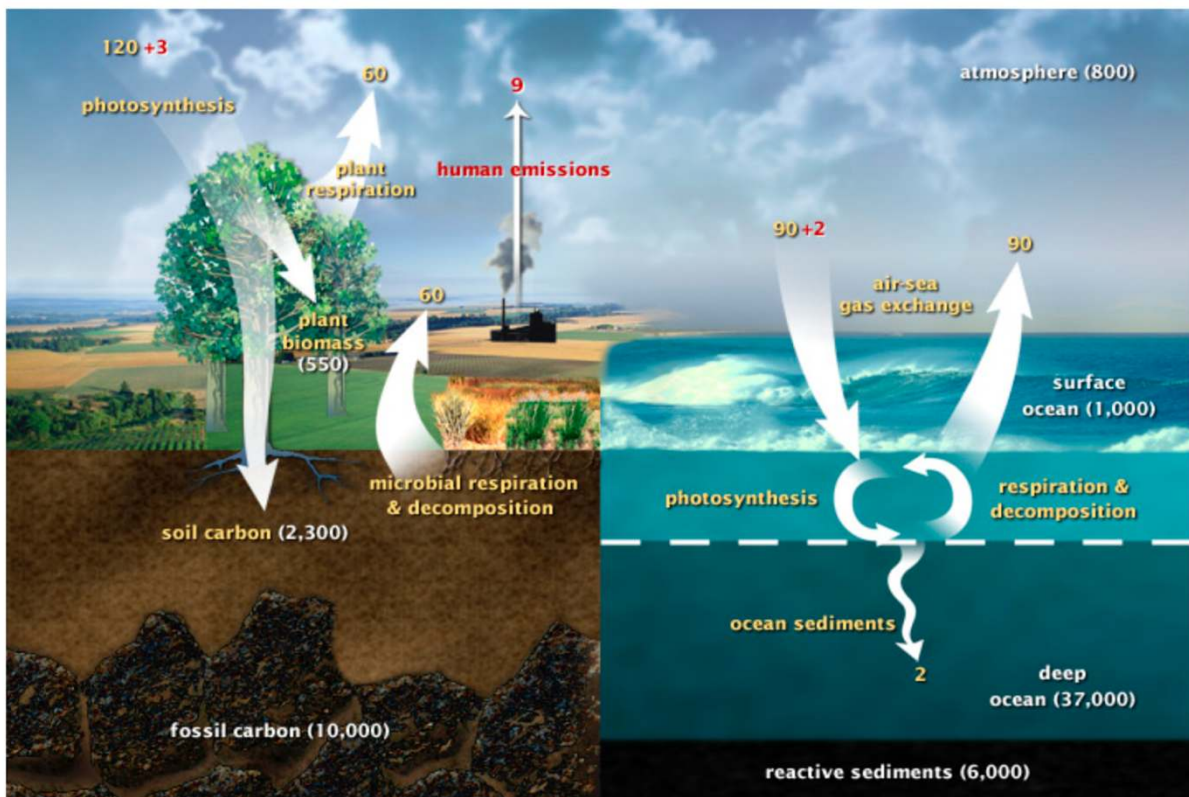
The solubility pump is however incredibly important due to its role in sequestering CO_2 in the deep ocean (which as you saw, is the largest store of carbon). We've mentioned that dissolution of CO_2 in sea-water is dependent on temperature. The colder the water is, the more CO_2 the water is able to hold. Unsurprisingly then, the polar regions are where the ocean sucks up much of the carbon that goes into the ocean each year. This also happens to be the regions in which deep water is formed, which then takes this carbon-rich water and pumps it into the deep ocean. This process has been integral in tempering the effects of [anthropogenic](#) emissions on the atmosphere.

In addition to the solubility pump, which physically transports [DIC](#) into the ocean's interior, there is also the biological pump. It is a little-known fact that at least half of the oxygen we breathe is actually produced in the ocean, and instead of by trees, by [phytoplankton](#). Much like trees, [phytoplankton](#) in the surface ocean take up [carbon dioxide](#) and sunlight in order to grow and create oxygen as a byproduct. Zooplankton then eats this [phytoplankton](#) before dying or being eaten themselves. Dead plankton (their organic matter retaining the carbon they removed from the ocean/atmosphere) are denser than surrounding seawater and so begin to sink. Over time, this rain of [organic carbon](#) exports carbon from the surface into the deep ocean, where it can remain for hundreds or thousands of years. This is the ocean biological pump – the biological fixation of carbon by [phytoplankton](#) before being exported to the deep ocean by predation or mortality (see the figure below).



Anthropogenic Emissions

Now that we have a rudimentary understanding of the [carbon cycle](#), let's return to the figure below.



Note that human emissions are relatively small compared to the fluxes of carbon in and out of the ocean and land. How can they matter so much? Annual exchange of carbon from the ocean to the atmosphere alone is on the order of 90 [gigatonnes](#) of carbon - 10 times greater than [anthropogenic](#) emissions. The important thing to note here is that the ocean also takes up about the same amount - the system is close to a state of equilibrium. It's not that our emissions are large compared to existing exchanges within the [carbon cycle](#), it's that the emissions are occurring too quickly for it to compensate. The stock of carbon in the atmosphere increases because the flow in now exceeds the flow out, even though the oceans and [biosphere](#) are partially compensating for this (the "+2" and "+3" in red in the figure above).

If we stop emitting carbon into the atmosphere, the [carbon cycle](#) will eventually reach a new equilibrium where inflow and outflow from each sink is once again roughly the same, but due to the time-scales the ocean [carbon cycle](#) operates on, this could take decades to centuries. Once this equilibrium is reached, the [CO₂](#) concentration in the atmosphere will stabilise and not increase anymore. It will stabilise at a much higher level than where we started from before the industrial revolution.

Lesson5:

1 - Observing the climate

Introduction

In order for us to understand how the climate has changed, first, we need to be able to measure it. Important climate variables such as temperature, rainfall, humidity and wind, as well as less well-known but equally important variables, are measured across the globe using a range of different methods. But how do we measure this data, and how do we know if the measurements we read, or the trends we calculate from them, are accurate? This lesson will describe the different ways that the state of the atmosphere is measured, from ground observations to satellites, as well as how scientists handle issues of data quality and changes in how observations are made.

The history of climate measurement

Aristotle was perhaps the first to document weather observations with his description of the hydrological cycle. It would be a long time, however, before we found a way to quantitatively make measurements of the weather and climate. The first equipment to measure weather variables was created in the 16th and 17th centuries, with early thermometers invented by scientists such as Galileo and Santorio. Early thermometers often had no scale at all, with scales becoming more common in the 18th and 19th centuries which resulted in the first ever daily weather forecast in 1860. The [SI](#) unit for temperature, Celsius, is named after the Swedish astronomer Anders Celsius, who created it in 1742. Barometers, which measure the pressure of the atmosphere, were also invented during the mid-17th century.

The oldest instrumental temperature record in the world is the Central England Temperature dataset, which has monthly temperature data starting in 1659. The number of places recording temperature has increased through time, with temperature observations in Sydney starting in 1858. The bottom left panel in the figure below shows when different temperature records began. While difficult to see, before the mid-20th century, the majority of temperature measurements were being made in Europe and North America, with a handful in Australia.

During the 19th century, sailors also began using thermometers to measure sea surface temperatures along major shipping routes (see figure above, right). Over time, many more ways to observe the atmosphere have been invented. From the middle of the 20th century, weather balloons have been released into the atmosphere. These provide information about weather above the Earth's surface, including how winds and temperatures change with height. Weather [radars](#) were invented during the Second World War, and are increasingly used to provide detailed information about rain rates and wind speeds. Finally, both geostationary satellites (only look at one location) and polar-orbiting satellites (pass above both poles) have allowed us to monitor the climate of the whole world since the 1970s, with continual improvement in the resolution of observations and the range of data available.

How do we measure climate?

There are many other sources of global weather and climate data, beyond weather stations and the [gridded](#) datasets made from them. All have different benefits and disadvantages.

Documentary evidence

Historical documents such as newspaper articles can provide great information about past weather, particularly extreme events. These have been used to create databases of historical severe floods or droughts, to allow better assessment of long-term changes. However, news reports often exaggerate the severity or rarity of an event and need to be checked against other observational evidence as much as possible (remember when we talked about the problems with [experiential processing](#) in the psychology lecture?).

Satellites

Since the 1970s, satellites have enabled the production of global datasets that have climate data over previously unobserved places. These can produce very high resolution and spatially complete data for a large number of variables, including vegetation greenness, cloud cover, [land cover](#), dust storms, and bushfires.

However, satellite data is only available since the 1970s, with many instruments only available for ten or twenty years. Changing instruments and satellites mean the data needs to be very carefully calibrated to create a more consistent dataset - otherwise, the dataset will incorporate [bias error](#).

Satellites do not directly measure the same property as ground data. For instance, a satellite measurement of temperature is typically either the "skin temperature" of the earth, equivalent to holding a thermometer on the hot ground instead of inside a standard [Stevenson screen](#), or rely on a very complex set of equations to calculate air temperature from the amount of radiation the satellite receives at different [wavelengths](#). This makes the data difficult to compare with ground-based measurements.

Radars

[Radars](#) are increasingly used for short-term forecasting and observations of weather, such as severe thunderstorms. These can provide rain rates at very high spatial and temporal resolutions. However, as with satellites, they do not directly measure rainfall. Instead, they measure the [reflectivity](#) of water droplets hundreds of metres to kilometres above the surface, which is converted into rain rates using complex equations. This also needs to be carefully calibrated against ground rainfall observations, and typically have data for less than twenty years.

Paleoclimate "[proxy](#)" records

[Proxies](#) are time series of observations that have strong relationships with meteorological variables, such as temperature or rainfall, and can be used to provide estimates of global climate thousands or millions of years ago. These include tree rings (see image), ice cores, stalagmites, or lake sediments. They provide much longer records than instrumental data, allowing us to put recent changes into historical context. However, they are not a direct measurement and have very low spatial and temporal resolutions by comparison (think one measurement per month/ year/ century as opposed to one measurement per hour/ day for modern day instrumental data).

Reanalyses

A [reanalysis](#) takes observational datasets, including satellite and station data, and assimilates them into a weather forecasting model. This provides global, physically-consistent fields of large numbers of climate variables, derived using the same physical equations and processes used for weather forecasting. Many modern [reanalyses](#) have resolutions of close to 50 km and 1 hour, allowing scientists to analyse data on a large range of scales.

However, [reanalyses](#) can often have inhomogeneities themselves due to the changing availability of data, such as when satellite data became available in the 1970s. And, while they can provide climate analyses in areas with no station data, this is not necessarily the "truth", with different [reanalyses](#) often disagreeing with each other.

The problem of observational errors (Sampling error)

There are three main sources of [uncertainty](#) in global climate data: **sampling error**, **bias error**, and **observational error**. In this section, we will use temperature data as an example; however, these uncertainties are present in many different climate variables.

Sampling error

[Sampling error](#) refers to errors in global climate due to the uneven distribution of observations in space and time.

In a perfect world, temperature would be measured at every point on the globe, and at every point in time. In the real world, climate data is concentrated in places like Europe, North America, and Australia, with relatively few stations in uninhabited areas such as the Australian outback or close to the poles. This is even truer when trying to look at historical changes, with many stations only beginning to operate during the 1950s. Before this, much of the temperature data was coming almost exclusively out of North America and Europe, as mentioned previously.

Large amounts of weather and climate data also exist as paper records that have never been digitised (see image below), especially prior to the 1900s. Other data has been lost or degraded over time. These also hinder our ability to develop good global datasets.

Sea surface temperatures ([SSTs](#)), prior to the advent of satellites, were mostly measured along global shipping routes, meaning many areas of the global oceans have almost no measurements prior to the beginning of the satellite era in the 1970s. The [correlations](#) between temperature [anomalies](#) at different stations can still be very strong, even when they're thousands of kilometres apart. Because of this, even a relatively small number of stations around the globe can be enough to estimate global temperatures. This means that the assessments of global annual temperature produced by organisations like NOAA, NASA and the UK MetOffice are generally very similar, with relatively small errors. The small differences between these different datasets are generally due to differences in how they handle the problems of missing data in areas such as the Arctic.

Bias error

The second source of [uncertainty](#) comes from methodological changes in how climate variables are measured, which can lead to biases in the data. This can happen as a change over time, such as in the figure below where several different measurement techniques all

measuring [SSTs](#) are actually measuring slightly different things. These differences are important and need to be accounted for, otherwise, you may end up with strange trends or breakpoints in your data which are not due to changes in the variable being measured.

A great example is a clear decrease in the raw global sea surface temperatures ([SSTs](#)) data in 1945. There is no drop in land surface temperatures at the same time, and no obvious volcanic eruption to cause a global cooling.

A detailed study in 2008 found that this temperature change was actually due to a change in observation practices. At the end of World War II, there was a change in the [SST](#) dataset from mostly US to mostly UK ships (see figure below). As it turns out, these two countries measured temperatures slightly differently - while the American ships measured the temperature of water brought into the ship's engine room, the British ships used an old-fashioned bucket to collect water over the side of the boat to measure.

A 0.3°C difference isn't much in the course of day-to-day measurements. But it makes a big difference to the assessment of long-term trends, with global temperatures rising by about 1°C over the past century.

Similar small but not insignificant changes can arise across the whole network from things like changing from Fahrenheit to Celsius, or changing the measurement time to be 9 am instead of midnight, or the introduction of daylight savings. And it's not just station data - slow drifts in the orbits of satellites need to be carefully corrected for because if a satellite gets slightly slower it will change what time of day temperature in a certain place is measured.

Observational (station) error

The third source of [uncertainty](#) is [uncertainty](#) due to the observations themselves. We call these station or observational errors.

The [World Meteorological Organisation](#) ([WMO](#)) sets global standards for weather observations (temperature, precipitation etc). Stations that satisfy these standards are more likely to have high-quality data that is free from errors which might hide or exaggerate changes in local climate. These stations can be used for assessing long-term changes.

Ideally, an observation site should be:

1. Representative of its surroundings
2. Not on steeply sloping ground or a hollow
3. On watered, mowed grass, or a ground cover representative of the area
4. In an enclosure of at least 10 × 7 m, and well away from obstructions
5. Have good metadata of how the site is arranged, what instruments it uses, and any changes over time (example in figure below right)

The [WMO](#) also sets standards for individual measurements, such as instrument accuracy, time of observation, or how the variable is measured. For instance, temperature data is recommended to be measured inside a [Stevenson Screen](#) (the white box in the photo above, left).

However, not all measurement sites meet the [WMO](#) standards. Data can also be affected by a number of quality issues including:

- Human error, e.g. 24°C accidentally recorded as 42°C or 2.4°C)
- Instrumental error, e.g. a rain gauge overflowing during extreme rainfall
- Gaps in the record, that can influence monthly or annual totals
- Observer behaviour, e.g. observers taking Sunday off leading to [anomalously](#) high rainfall totals on Mondays and rain rarely recorded on Sundays (this is surprisingly common)

All of these issues need to be carefully identified and corrected in the raw data before they can be used for analysing global climate or trends.

The quality or nature of observations can also change over time:

- Changes to where the measurements are taken
- Changes in the instrument used
- Changes in observing practice, such as the time of observation
- Changes in environment, such as from watered grass to sand

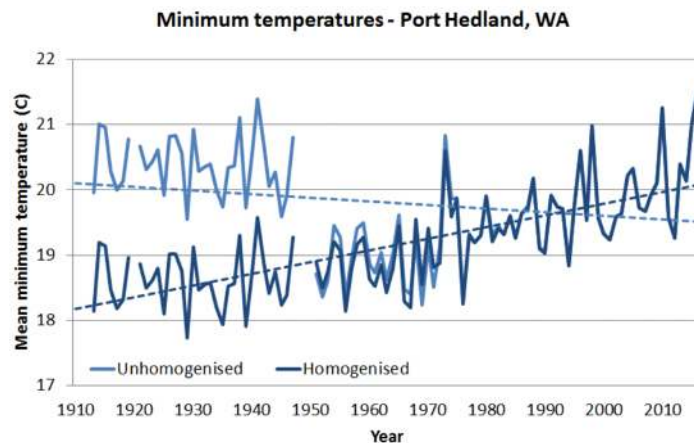
Problems with data quality, or changes in how data is observed, can lead to big changes in the data that don't reflect actual changes in the surrounding climate. We call these inhomogeneities.

Inhomogeneities

Earlier, we mentioned the concept of inhomogeneities - changes in the data at a station that are due to changes in the site environment or observing practices, rather than changes in the climate of the area.

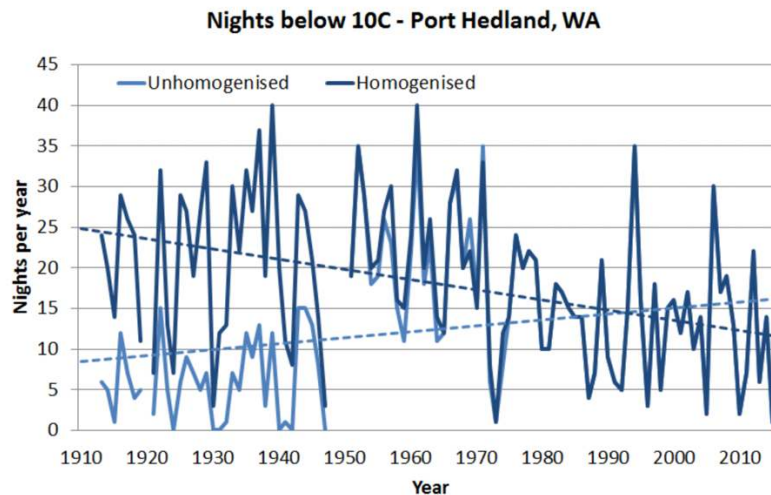
The figure below shows an example of minimum temperatures at Port Hedland in Western Australia. In 1948, observations moved from the Post Office, in the middle of town, to the Airport, 9 km away. The new site has much colder temperatures overnight, with an average minimum temperature about 2°C colder.

If you combine the two stations together and calculate a trend, it looks like the temperature has decreased strongly over time. But this trend is artificial - it's due to changes in the measurements, not changes in the climate. In fact, if you account for the change in location, temperatures have been warming since the 1950s, with little change before then.



The raw (light blue) & homogenised (dark blue) data for T_{min} and the trends for each. We can see that the break in 1948 (pre-1948 in town, post-1948 at airport) results in a large inhomogeneity between the two stations.

Inhomogeneities have a particularly big effect on extreme events. At the same station, the 2°C change in average minimum temperature in 1949 resulted in the number of nights below 10°C jumping from about five times per year to about twenty times (figure below).

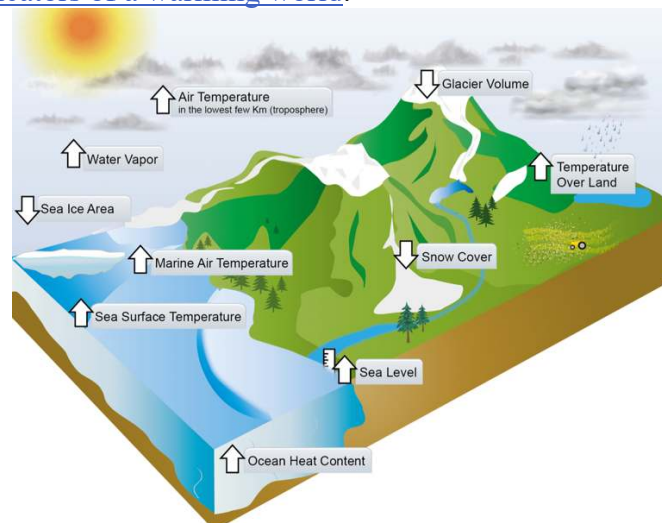


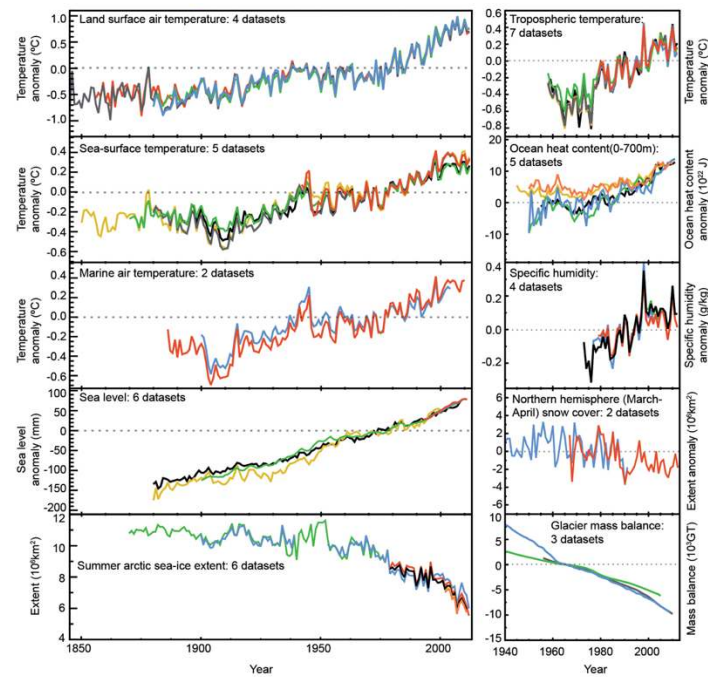
Not all inhomogeneities are this big - some are just a few tenths of a degree, like the changes in sea surface temperature data. This means we need sophisticated statistical techniques to test whether station data are homogeneous or not, and correct for these changes, before adding them to global datasets.

How can we be sure the world is warming?

We have a large understanding of the different sources of [uncertainty](#) in global data. This allows us to correct for some problems, like inhomogeneity, and calculate the magnitude of other uncertainties, like the spatial spread of data. This means we can calculate global average temperatures with [uncertainty](#) ranges, to accurately assess long-term changes. Globally, we now have about 160 years of reliable estimates, with some regions having more than 300 years of data.

There are also many different lines of evidence that confirm the same result. For instance, surface and satellite temperature trends broadly agree, and land and sea temperatures show consistent upward trends. Other biophysical impacts are also consistent with global temperature observations, such as melting ice and changes to the flowering of plants. The consistency across all the different types of data is a strong support for ongoing, global temperature increases. If you'd like to know more, have a read of this short article which discusses the [10 indicators of a warming world](#).





The indicators of a warming world (IPCC, AR5 WG1)

2 - Climate extremes

Introduction

Fires, floods, droughts, heatwaves: extreme events can cause major damage to life, property, and ecosystems. It is often said that the impacts of climate change will be felt most clearly through changes in the nature or [frequency](#) of extreme events.

When compared to changes in the average climate, the field of climate extremes is relatively new, with information about climate extremes first included in [IPCC](#) reports in 2001. This is a growing field of research, both in Australia and around the world.

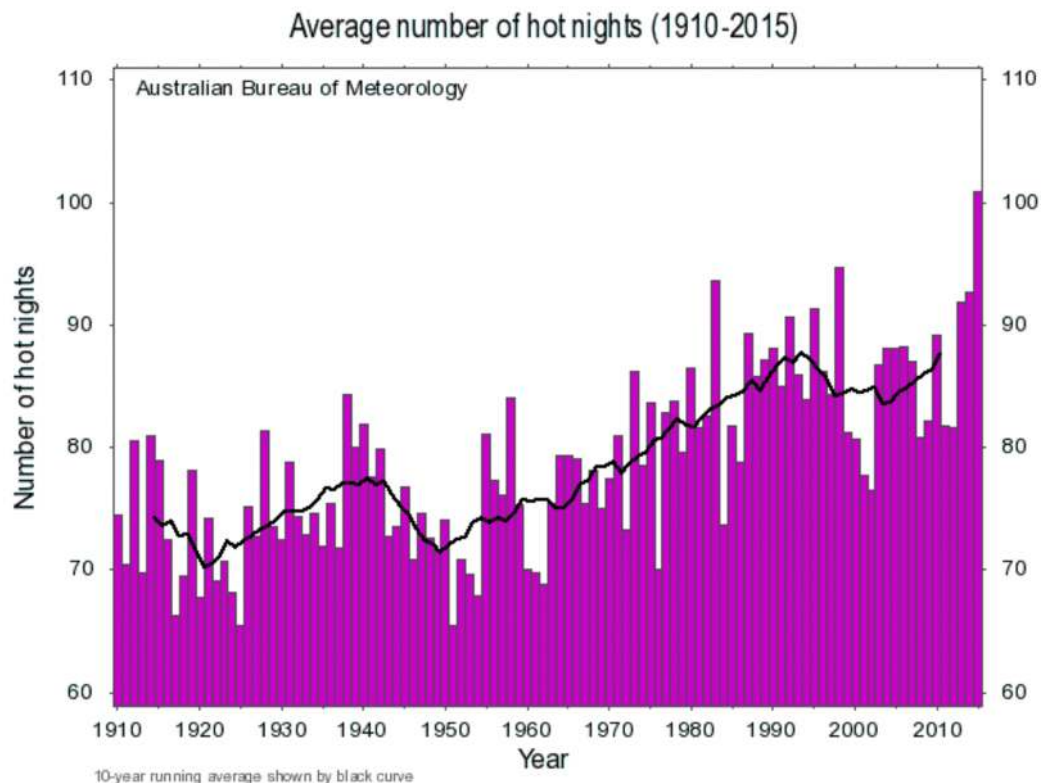
This lesson will provide an introduction to the science of climate extremes. It will discuss a couple of ways that extreme events can be defined and the latest understanding of how extreme events have changed over the past century.

What is a climate extreme?

"Extreme" can have a couple of different meanings. First, we have a sort of 'top-down' definition that looks at these events from a statistical perspective. To be specific, if during the course of a normal year, an event such as a record-breaking heatwave has a 1% chance of occurring, then we would say this is an extreme event as it is very unlikely. Events like this can be record floods, droughts, heatwaves and any other climate phenomena that are uncommon in their occurrence or severity. Using this definition we are essentially looking at extreme values of a climate variable (temperature, precipitation, etc).

The second way of thinking about extremes is a 'bottom up' approach. This focuses on the impacts of the event on human lives, infrastructure or ecosystems. A good example here is bushfires in Australia or hurricanes in the United States. While these two things occur regularly each year, their impact on both us and the environment can be catastrophic making them extreme in this sense.

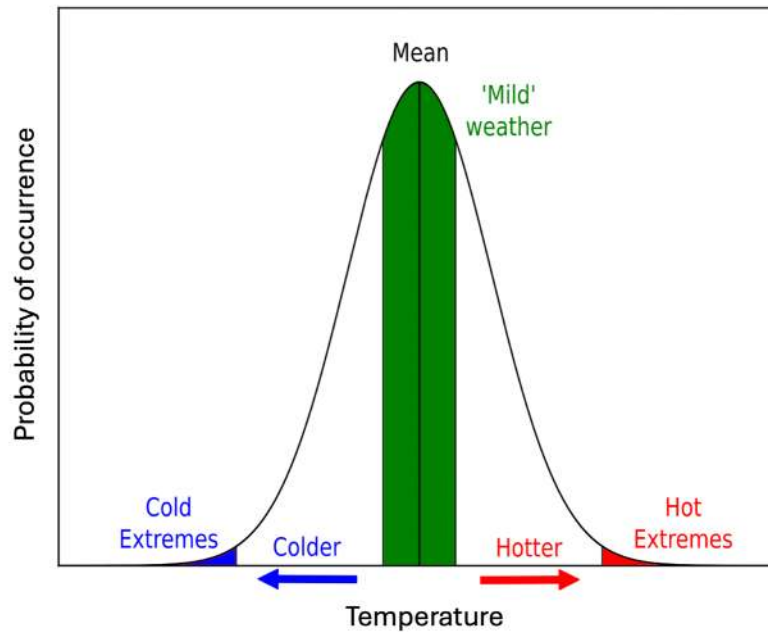
Of course, there is plenty of overlap between these two types of extremes, particularly on the wet end of the spectrum, but it provides a useful working definition. In this lesson, we will focus on four different extremes: heatwaves and hot extremes as well as heavy rainfall and tropical cyclones. We'll look at whether or not these events are changing in some way as the world warms and how confident we are in these changes occurring.



A brief statistics refresher

When people talk about the climate of a location, they often think about the average temperature or rainfall. But climate can be also be thought of as the possible range of some climate variable at the location - for this part, let's use temperature as an example. In a tropical climate, there is very little [variability](#) in the temperature from day to day or even season to season (hopefully you know why now!). In comparison, a place like Sydney or Melbourne can have a 10°C or even 20°C temperature swing between one day and the next when a cold front comes through. So the climate is more than just the mean - it's the range of temperatures, including the extremes.

We can visualise the whole range of temperatures at a location as some sort of "probability density function" (PDF). Looked at in this way, 'extremes' are the values that fall into the 'tails' - or either end - of the distribution, above (or below) some sort of threshold. See the example below.



An example of a normal distribution. Temperature, the value we want to know about, is on the X-axis, while the probability of that temperature occurring on a particular day is on the Y-axis. We can see that normal 'milder' temperatures occur most often. As the curve is symmetrical (its shape), the mean lies in the middle of all temperature values.

Extreme temperatures, by the statistical definition, do not occur very often, which is why they are in the tails of the distribution.

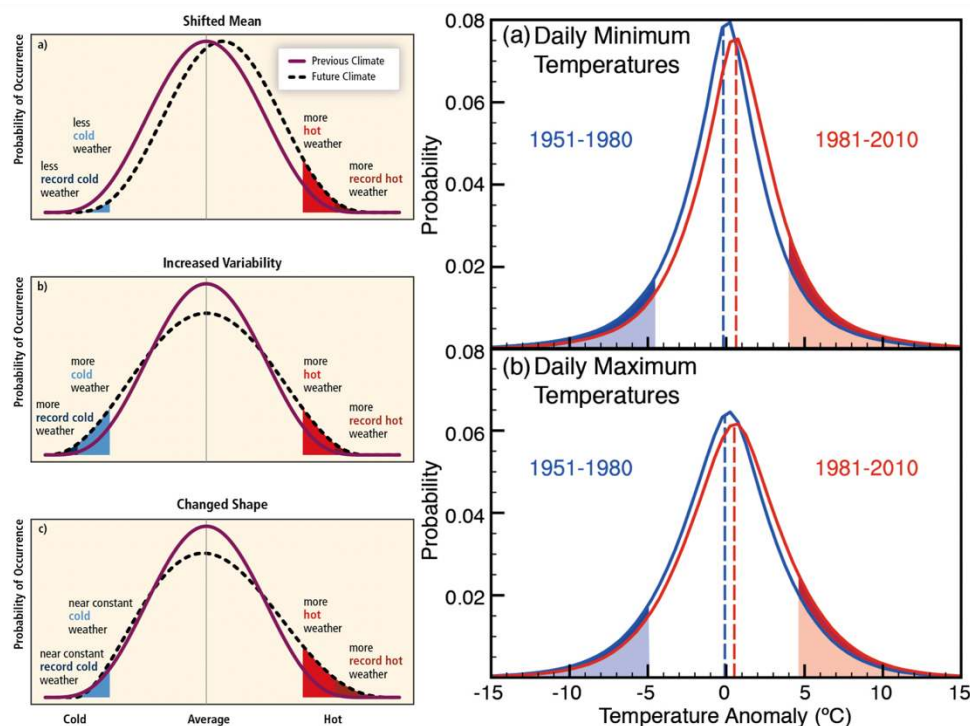
Statistics is a complex subject, and there are a lot of complicated statistical approaches to assessing extremes. We won't discuss them in great detail in this course, but it's useful to be aware of a few concepts:

- The normal state of the climate can be described using the mean (average) of a variable, or by the median (the value that exactly 50% of observations are above and 50% below).
- The [variability](#) of the climate is reflected in how wide the PDF is. A region with low [climate variability](#) would be somewhere like the [tropics](#) which are roughly the same temperature all year round. Somewhere with high [variability](#) would be, for example, parts of Europe or the United States which are hot in the summer, but cold and often snowy in the winter.
- The "shape" or "skewness" of a distribution describes whether the PDF is symmetrical, or whether it is skewed more towards high or low values. Variables like temperature are usually pretty symmetrically distributed, but something like rainfall may be a lot more highly skewed, with most days having relatively low rainfall but some very wet days. For instance, in Alice Springs rain only falls on 43 days per year, and about 80% of those days have less than 10 mm of rain. However, there are some days that have more than 100 mm of rain, including 204.8 mm on one day in 1988.
- The " X^{th} [percentile](#)" is the value that exactly X% of data falls below. So, the 5th [percentile](#) of temperature is the value that 5% of days are cooler than and 95% of days are warmer than, while the 95th [percentile](#) is the temperature that only 5% of days are warmer than.

In theory, changes to any or all of these characteristics of the climate can result in a change in the number of extremes. For instance, a simple increase in the mean temperature at a location would result in a lot more days that were very warm relative to the previous climate, and fewer days that were very cold (figure below left, panel (a)).

Even without a change in the average temperature, a change in the [variability](#) would result in a change in the [frequency](#) of extremes (figure below left, panel (b)). By stretching the distribution, higher [variability](#) would mean both hotter and colder extremes. A change in the shape or 'skewness' of the distribution could also influence the relative [frequency](#) of hot and cold extremes (figure below left, panel (c)).

So far, most of the changes we've observed in climate extremes on a global scale are driven by increases in the mean temperature. Between the period 1951-1980 and 1981-2010, the average minimum temperature increased by 0.8°C (figure below right, panel (a)). This resulted in a decrease in the [frequency](#) of very cold nights, and about a 40% increase in very warm nights.



(Left) Changes in the probability of climate extremes as the climate changes (IPCC SREX) and (right) observed changes in minimum and maximum daily temperatures (IPCC AR5, WG1)

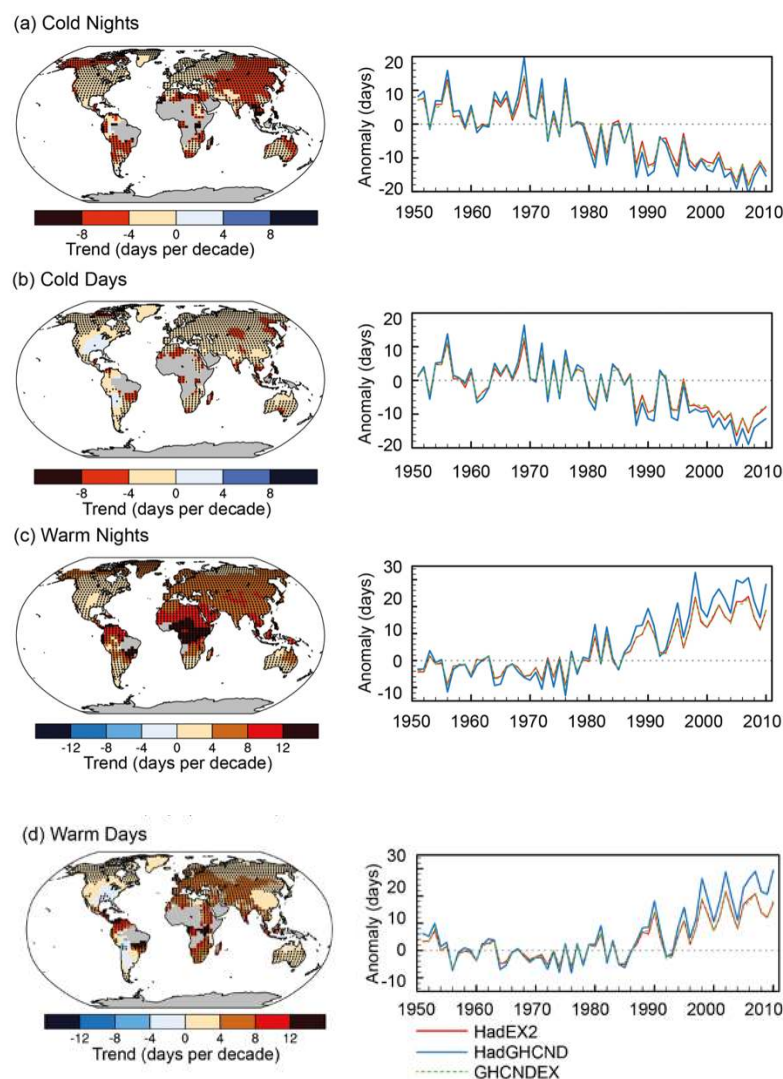
Hot days and cold nights

Following on from the theoretical shifts in temperature distributions you saw earlier, given that the global mean temperature has been rising, we would also expect to see an increase in the numbers of hot days and nights, and a decrease in the number of cold extremes (see figure below). Globally, that is exactly what we see. The [IPCC](#)'s Special Report on Managing the Risks of Extreme Events and Disasters to Advance Climate Change Adaptation (SREX) in 2012 reported that:

"It is **very likely** that there has been an overall decrease in the number of cold days and nights, and an overall increase in the number of warm days and nights, at the global scale, that is, for most land areas with sufficient data. (**high confidence**)"

The [IPCC](#) uses [calibrated language](#) when talking about observed and future projections, to distinguish between confidence in our scientific understanding and the likelihood of specific results. So this statement means that there is a greater than 90% chance that we have already observed increases in hot weather and decreases in cool weather across the globe over the last century.

The magnitude and confidence in observed trends varies across the globe. For instance, across much of the globe, there is high confidence that the [frequency](#) of warm nights has increased and the number of cool nights has decreased, with largest trends in Asia. There is also high confidence that warm days have increased and cold days have decreased over most of Europe and Australia, but there are some areas like the eastern USA that have seen either no change or a decrease in the number of warm days. Confidence is lower in areas such as Africa and South America where the quality and availability of historical climate data is lower.



Trends in warm and cold days and nights over the period 1950-2010 (IPCC, AR5 WG1)

In Australia, we have observed a significant increase in the [frequency](#) of extreme heat in recent years, with many severe heat events breaking long-term records since 2009. You can read summaries of many of these events at the [Bureau of Meteorology](#).

Heatwaves

One hot day can be unpleasant, and some animals can't tolerate temperatures above a certain temperature threshold. But the biggest health impacts tend to happen when we experience an extended period of warm days, especially when the nights stay warm and the body isn't able to cool off. We call an extended period of very hot weather a heatwave. Heatwaves are the most deadly of any climate extreme, with more than 70,000 people dying during the European heatwave of 2003.

Just like any other type of extreme, the exact definition of a heatwave varies from place to place. In a colder country like Denmark, a heatwave is considered a spell of three or more days with temperatures above 28°C. In Adelaide, daytime temperatures are that warm throughout half the summer, so a heatwave would instead mean temperatures above 35°C or 40°C. In comparison, temperatures don't vary as much from day to day in the [tropics](#), where the largest health impacts may instead be seen during periods of very high humidity.

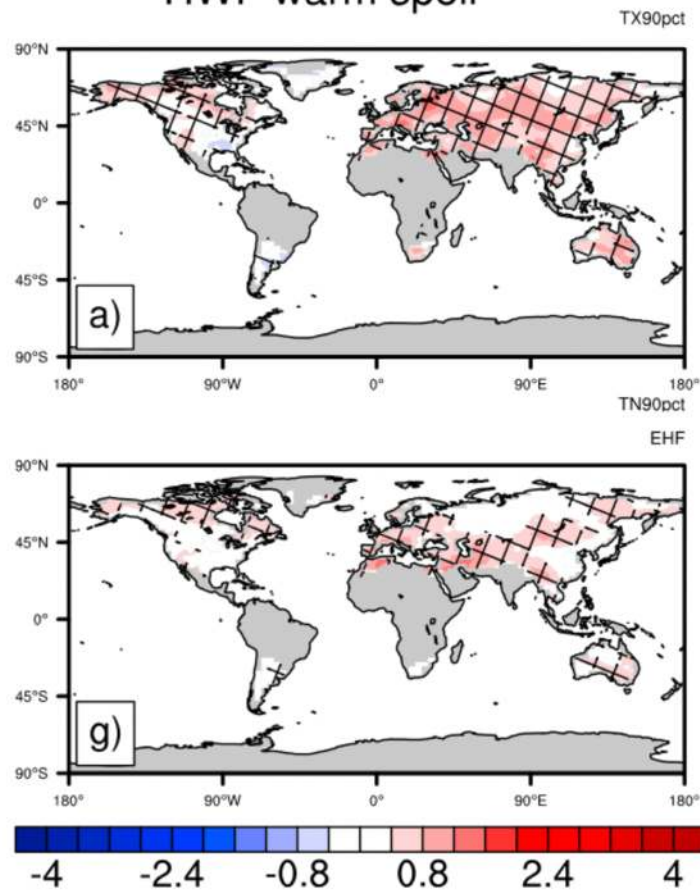
A consistent definition of a heatwave is "A prolonged period of excessively hot weather, which may be accompanied by high humidity". "Excessively hot" is relative to what is normal for a given location, so is often defined by a threshold temperature such as the 90th or 95th [percentile](#). For instance, the "Warm Spell Duration Index" in the ETCCDI set of standard indices looks at periods of at least 6 days exceeding the 90th [percentile](#), while the "Extreme Heat Index" used by the Bureau of Meteorology looks at a period of at least three days with mean temperatures above the 95th [percentile](#).

"Warm spells", periods of [anomalously](#) warm temperatures in the cooler half of the year, can also have large impacts (and have been increasing, see figure below), but here we will focus on the severe heatwaves that occur during the summer months.

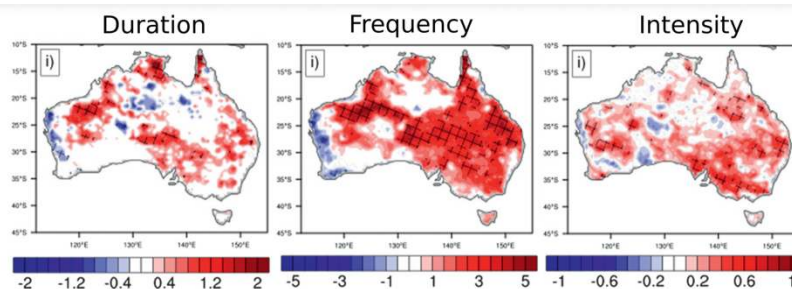
As with extreme heat, the [IPCC](#) SREX report states that "Globally, in many (but not all) regions with sufficient data there is **medium confidence** that the length or number of warm spells or heat waves has increased since the middle of the 20th century."

This trend is most clear in Europe and Asia but is sensitive to the way we define a heatwave. In Australia, it is likely that the [frequency](#) of heatwaves has increased across most of the country, with less certain changes in their intensity or duration.

HWF warm spell

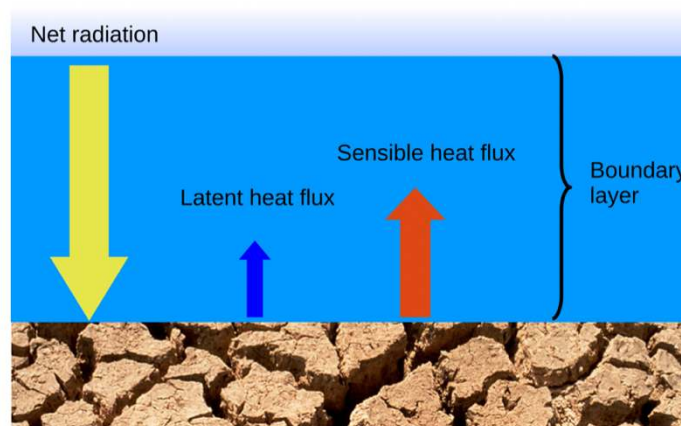
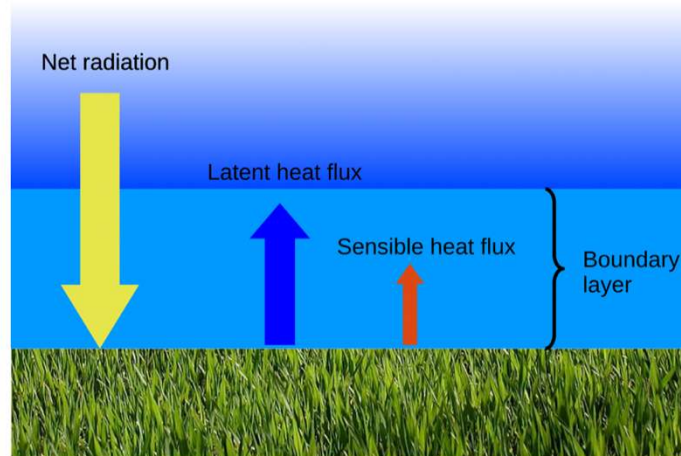


Trends in warm spells since 1950 (percentage of days/ season/ decade) adapted from [Perkins et al. 2012](#)



Trends in the duration (days/decade), [frequency](#) (days/decade) and intensity (°C/decade) of heatwaves in Australia (adapted from [Perkins and Alexander, 2013](#))

The analysis of changes in heatwaves and hot extremes is complicated by the interaction between heat and rainfall. For instance, very dry soils can increase the likelihood of severe heat, by decreasing the amount of evapotranspiration that occurs. This means that some of the incoming solar radiation being used to evaporate water, it is instead converted into heat.



When soil moisture is low, the **sensible heat** flux is greater than the **latent heat** flux, exacerbating the heatwave.

Extreme rainfall

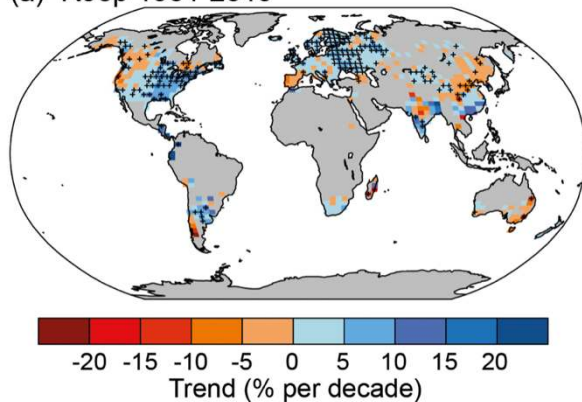
The amount of [water vapour](#) in the atmosphere has a strong relationship with temperature. For every 1°C rise in temperature, the total moisture levels are expected to increase by 7%.

This does not necessarily mean an increase in the total amount of precipitation, as rainfall is influenced by a number of factors such as changes to the types of weather systems that are common in a particular location. However, if all else is equal, storms that occur in a warmer atmosphere are expected to have a higher potential maximum rain rate, and so we expect an increase in the [frequency](#) of rainfall extremes across the globe, even where total rainfall decreases.

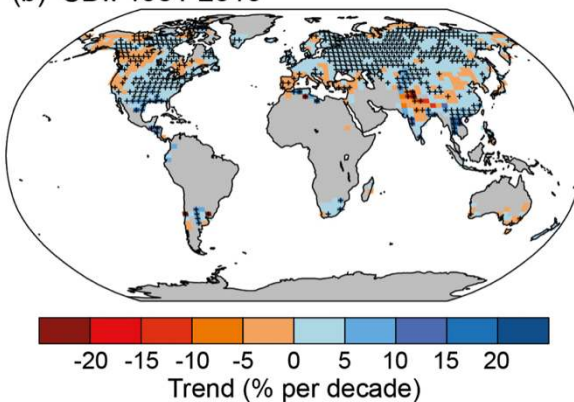
Rainfall is much more variable than temperature, and the observed changes so far are less significant. The [IPCC](#) SREX report states that "It is **likely (high confidence)** that there have been statistically significant increases in the number of heavy precipitation events in more regions than there have been statistically significant decreases, but there are strong regional and subregional variations in the trends."

Precipitation data is less reliable than for temperature, and it is particularly difficult to assess trends in very extreme rainfall (e.g. the wettest day per year) or sub-daily rainfall totals. However, there are significant increases in the proportion of rainfall that is due to very heavy rain days (above the 95th [percentile](#)) across much of Europe and North America.

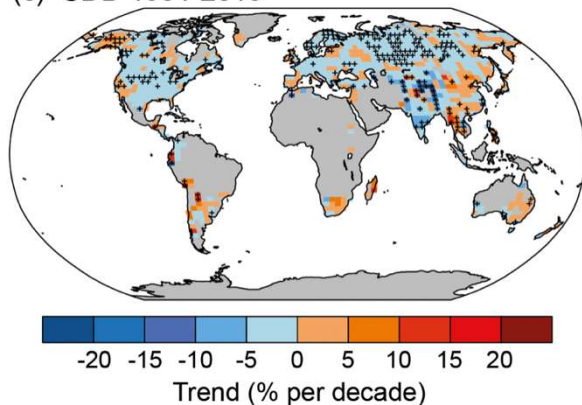
(a) R95p 1951-2010



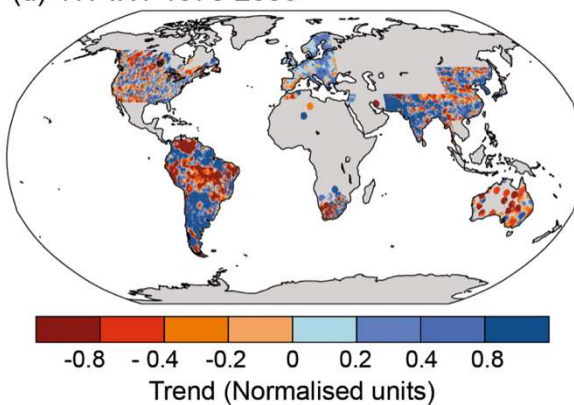
(b) SDII 1951-2010



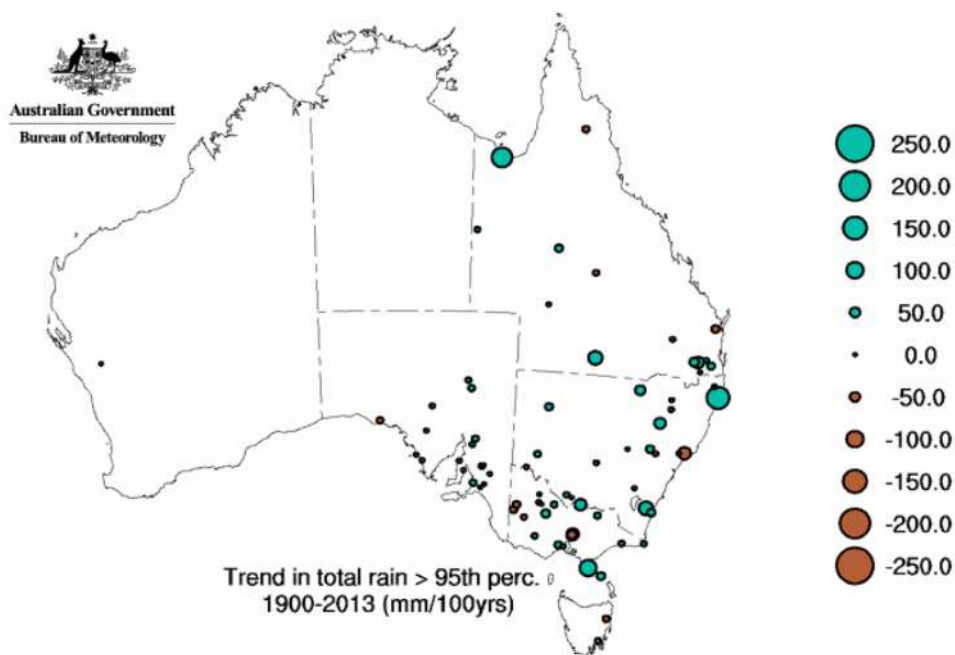
(c) CDD 1951-2010



(d) HY-INT 1976-2000



Trends in (a) annual amount of precipitation from days > 95th percentile (how much rain falls in heavy rainfall events); (b) daily precipitation intensity; (c) frequency of the annual maximum number of consecutive dry days and (d) hydroclimate intensity. '+' in panels (a)-(c) indicate where results are statistically significant at 90% level (IPCC AR5, WG1 - Ch.2).



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Trends in heavy precipitation over Australia from 1900-2013 (BoM).

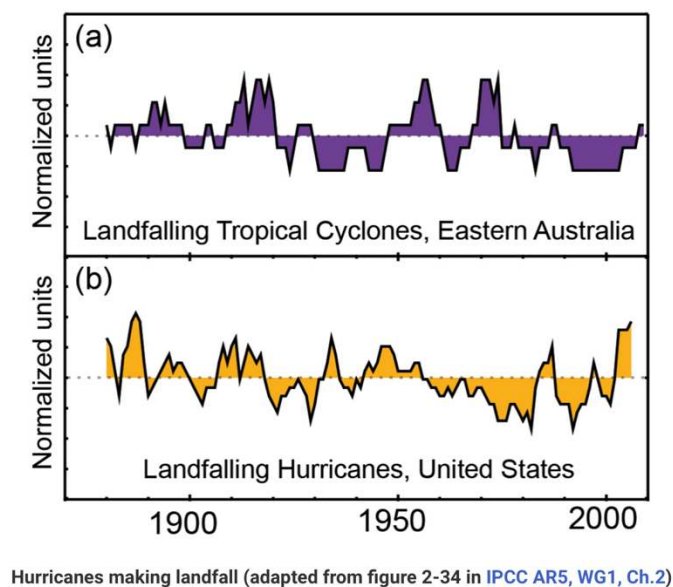
Tropical cyclones

Tropical Cyclones are not a "climate extreme" like the previous extremes we've looked at. Instead, they are complex weather systems that are characterised by their structure (such as having a defined "eye") and their maximum wind speed. Tropical Cyclones can cause very severe flooding and damage, due to their heavy rain, strong winds, and large waves/storm surge.

Typically around 13 cyclones form each year in the Australian region, with about six crossing the Australian coast. Australia usually has more tropical cyclones in [La Niña](#) years and lower frequencies during [El Niño](#).

Because of the lack of historical observations over the oceans, it is very difficult to robustly assess the total number of cyclones or their intensities before 1979. Taking into account the fact that many early cyclones will be missing from our records, it is unlikely that the total number of tropical cyclones has increased. The [IPCC](#) SREX states "There is **low confidence** that any observed long-term (i.e., 40 years or more) increases in tropical cyclone activity are robust, after accounting for past changes in observing capabilities."

In Australia, observations indicate that the total number of tropical cyclones has decreased since the 1970s, with no clear trends in the [frequency](#) of severe cyclones (category 3 or higher). Records from newspaper reports suggest a slight decrease in the [frequency](#) of cyclones that cross the Queensland coast. In comparison, there has been an increase in cyclone [frequency](#) observed since the 1970s in the North Atlantic (see figure below).



So, if we have low confidence in whether there have been any changes in tropical cyclones and extreme rainfall, does that mean we have nothing to worry about?

Not necessarily!

'Low confidence' neither implies nor excludes the possibility of changes in extremes, it just means we don't have the quantity or quality of data we need to assess them properly. This is particularly true for very variable events like cyclones or thunderstorms.

The full effect of climate change is also not being realised yet, with the large natural [variability](#) in extremes making it harder to detect any long-term changes. Paradoxically, we may be able to make more confident statements about changes in future extremes, as the global climate becomes further outside the envelope of historical [variability](#).

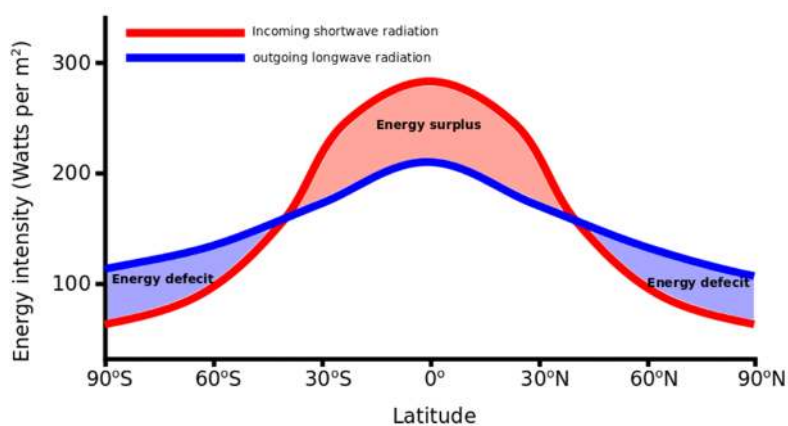
Lessons6:

1 - Atmospheric circulation

Why does the atmosphere move?

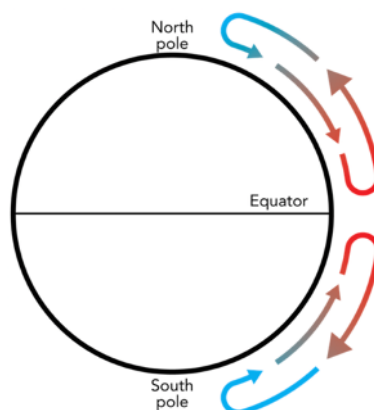
The atmosphere moves for a reason. If you remember back to the lesson '[Energy to Earth](#)', you learned that the intensity of the sun's incoming radiation is disproportionately strong near the low latitudes (the [tropics](#)), as you can see in the figure below. You also learned that the atmosphere (the [troposphere](#), more specifically) warms from **below** due to the [infrared radiation](#) emitted by the Earth. These two things have two consequences:

- 1 - The atmosphere is warmer at the equator than at the poles and
- 2 - The atmosphere is warmer at the surface than at higher altitudes.



The Earth balances incoming solar radiation with outgoing [infrared radiation](#). Note that there is a surplus of energy at the equator and a deficit at the poles. This uneven distribution of energy causes the atmosphere to move.

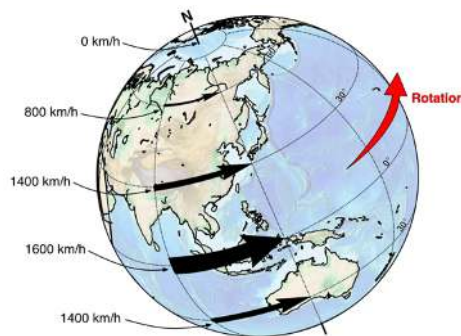
To examine the consequences of these things, let's imagine an idealised Earth - no land, no water just a smooth planet of dirt. The warm air near the equator wants to rise, as it is less dense than the air above it - this is known as [convection](#). But there is also a latitudinal [gradient](#) in temperature, so as it rises, it moves north and south towards the poles where the air is cooler and denser. At the surface, cold dense air from the poles rushes in to fill the gap left by the rising warm air at the [tropics](#). In this ideal scenario, we are left with the figure below - two '[convection](#)' cells, one in each hemisphere which act to equalise the energy imbalance observed in the figure above.



Hot air rises at the equator and moves towards the poles. There it cools, becomes denser again and moves back towards the equator. This is a [convection cell](#) (Image courtesy of [Byron Inouye](#)).

The Coriolis effect

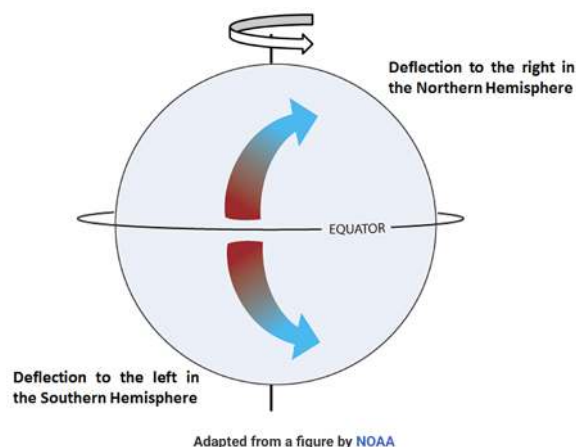
The idealised scenario we discussed on the previous page does not work for Earth, however. Things get complicated when we consider that our planet is rotating at a considerable speed, resulting in every moving object being affected by the [Coriolis force](#). Our planet is constantly rotating towards the east, however as the Earth is a sphere-like object, the speed at which we travel eastward depends on the latitude we are at. For example, if you look at the figure below the land at the equator is moving at roughly 1,600 km/h, while at the north or south poles the speed is necessarily zero.



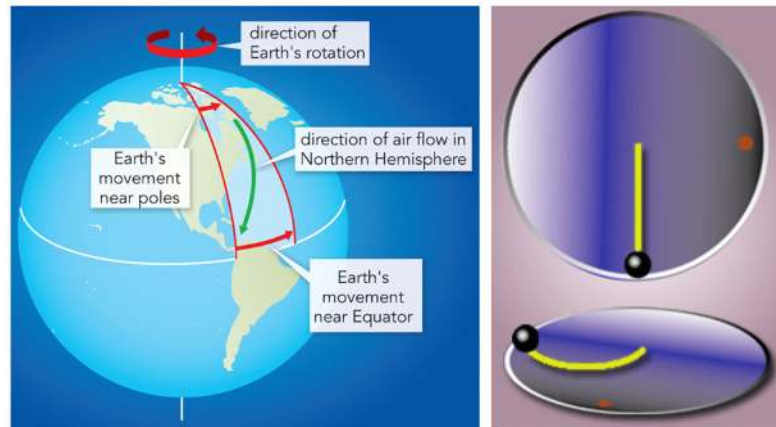
So how does this difference in eastward velocity with latitude affect the movement of our atmosphere and oceans? The key is understanding how the motion of an object changes relative to the motion of the Earth as it crosses from one latitude to another.

Let us imagine a rocket sitting at the equator. This rocket is travelling eastward with the Earth at 1,600 km/h so that relative to the Earth, it is simply sitting still. Next, we launch the rocket directly north - but remember, it still has an eastward velocity of 1,600 km/h that it had at the equator. As it continues north, the Earth spinning underneath it is going east less quickly than the missile (say, 1,400 km/h at 30°N as opposed to the 1,600 km/h of the missile) so to us, it appears as if the missile has begun to fly not only north, but also towards the east.

The same thing applies to the wind and the ocean; when wind blows or the ocean flows across latitudes, they will be deflected - to the right in the Northern Hemisphere and to the left in the Southern hemisphere (see figure below).



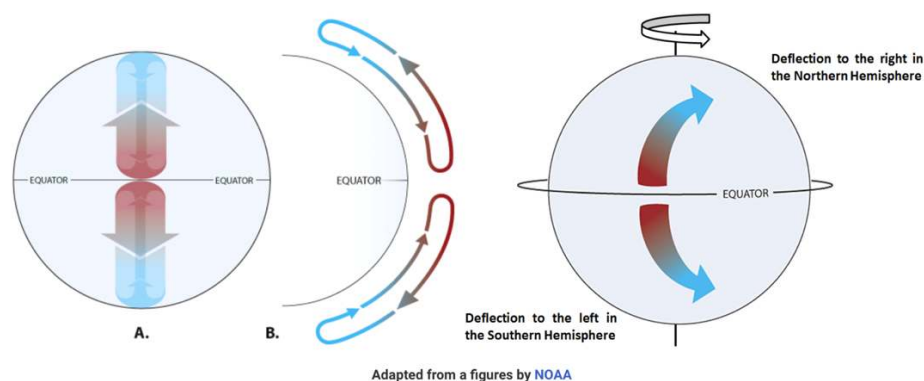
The figure below shows the opposite of our example above: air flowing from the poles to the equator. To re-iterate, the air at the poles has almost no eastward velocity, so as it moves south, the Earth is moving eastward at a far greater speed relative to the air. As a result, the air looks like it 'lags behind' the spin of the Earth, and bends towards the right as it travels south. Hopefully, from these explanations, it is clear that the Coriolis effect is not actually the result of a 'force' but simply a consequence of the difference in eastward velocities on our rotating planet.



(Left) Resulting movement of an object that flies from the pole to the equator (Image courtesy of [Byron Inouye](#)) and (Right) an example of this same scenario on a flat rotating disc (Image courtesy of [Hubi](#))

The large scale movement of the atmosphere

So now we know enough to put together how the large scale circulation of the atmosphere works. First, we have warm air rising at the equator and moving towards the colder poles (figure below left), however now we also know that as these air masses move across latitudes, they will be subjected to the coriolis effect and deflected to the right in the northern hemisphere and left in the southern hemisphere (figure below right).



The Hadley cells

This results in a series of atmospheric 'cells' - three in each hemisphere that form due to the combination of these forces. At the equator, from 0°- 30° N or S, we have the Hadley cells. These are the result of warm, moist equatorial air rising and moving towards the poles. The combination of cooling and deflection due to the coriolis effect results in the air descending

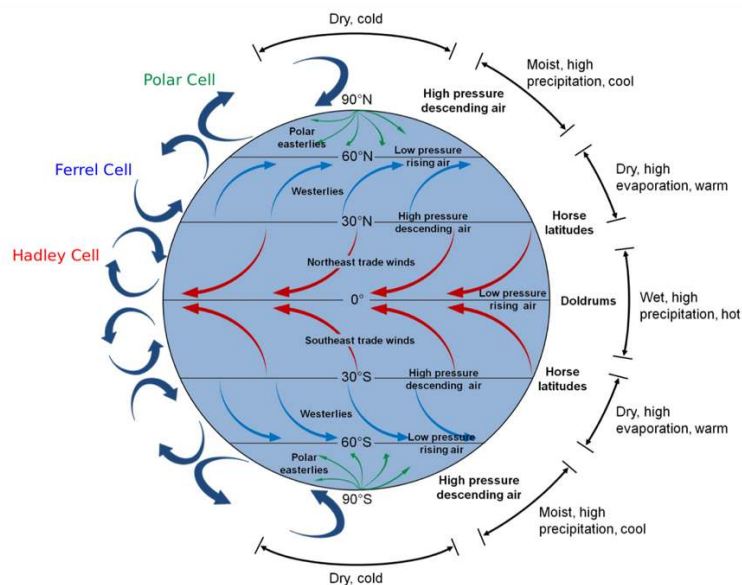
again around 30° N/S, now much cooler and dryer than it was (remember, cooler air cannot contain as much [water vapour](#)). This cool, dry surface air then moves back towards the equator as the [trade winds](#) - this closes the [convective](#) loop of the Hadley cells.

The Polar cells

The Polar cells between 60° and 90° latitude work in a similar way (see figure below). The air at 60° is more warm and moist when compared to the cold polar air at 90°. This allows another [convective](#) loop to form where air rises and moves towards the poles, losing moisture and heat along the way. Much like the Hadley cells, it then sinks towards the surface as drier, colder air and moves back towards lower latitudes, closing the loop.

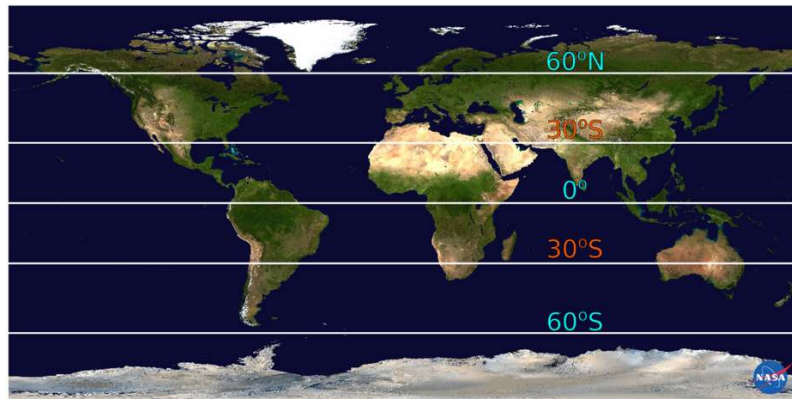
The Ferrel cells

The Ferrel cell is the final atmospheric circulation between 30° and 60° in latitude. Unlike the Hadley and Polar cells however, it is not driven by temperature differences and the associated [convective](#) processes. Instead, it can be thought of as a byproduct of the Hadley and Polar cells. Some of the air descending at 30°, instead of flowing back towards the equator, flows the other way towards the poles while at the same time, some of the air rising at 60°, instead of flowing towards the poles, instead heads towards the equator. The result of this is a third circulation pattern, weaker than the Polar and Hadley cells that sits between 30° and 60° latitude.



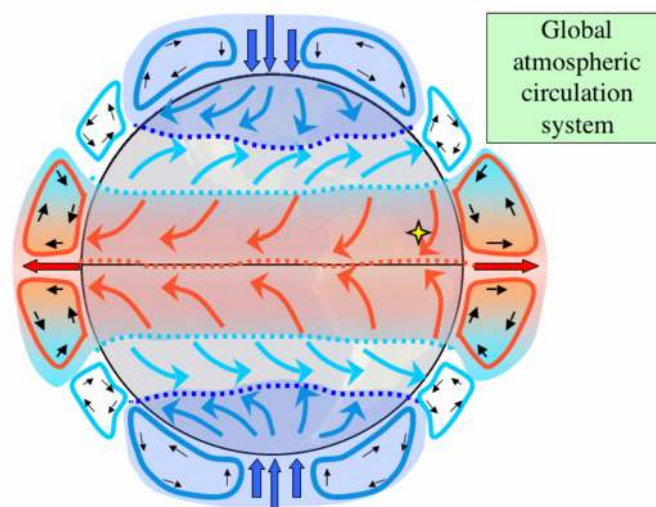
These circulation features have an enormous impact on where certain climates are found throughout the world. For example, at the equator and around 60° north and south where we find the ascending branches of the Hadley and Polar cells, we also find the parts of the Earth with the most rainfall (remember from previous lessons that the [convection](#) of warm moist air is how we get rain). If you have a look at the map below, you'll notice that all the world's largest rainforests are centered about the equator - the Amazon in South America, the rainforests of Indonesia and Papua New Guinea and the Congo Basin in Africa. Similarly, at 60°N we can see the boreal forests of North America and the forests of northern Eurasia.

On the other hand, at latitudes of 30° north or south where dry, cool air returns to the surface, we notice that this is where all the world's largest deserts lie such as the Atacama in South America, the Sahara in Africa and of course the Great Sandy Desert in Australia. With no large scale [convection](#) and therefore no clouds or rainfall, these latitudes are the driest on Earth.



A satellite photo of the Earth with latitude markers showing wet and dry regions. Photo courtesy of NASA.

You may also remember from the [Energy to Earth](#) lesson that seasonal changes in incoming sunlight shifted the [Intertropical Convergence Zone](#) north and south. Well, now you know that the [ITCZ](#) is simply the ascending branch of the Hadley cells and furthermore, all large scale circulation cells shift (at least a little) due to these seasonal changes in sunlight. The schematic below demonstrates this.



These circulation patterns shift with the seasons (Image courtesy of [Linsyl Urbano](#))

2 - Oceanic circulation

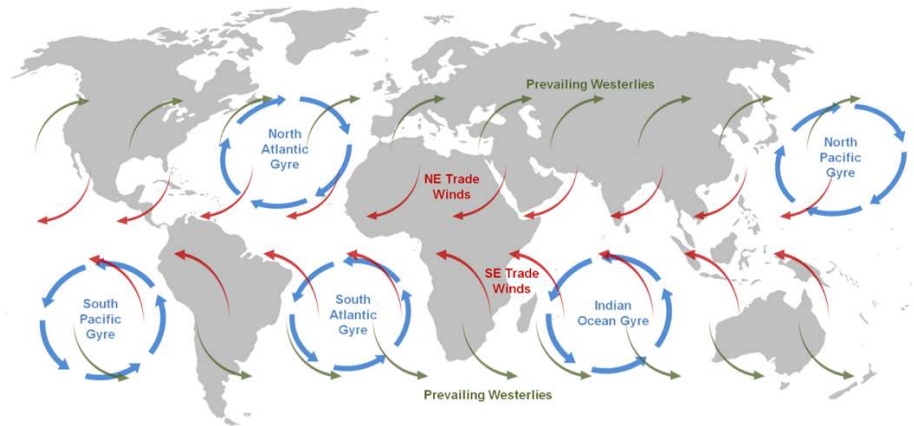
Why does the ocean move?

The ocean, unlike the atmosphere, moves for two different reasons:

1. Wind blowing on the surface and
2. Differences in density

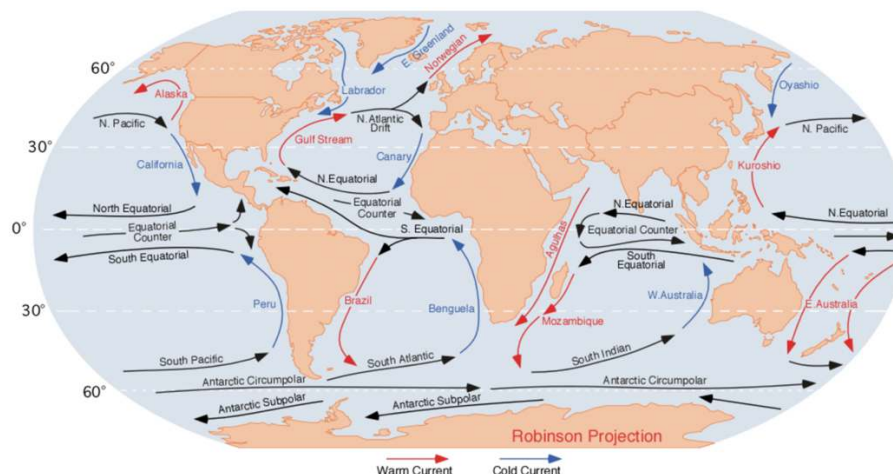
(Tides also impact ocean circulation but we will not discuss them in this lesson).

To begin with, we'll start with the wind-driven surface circulation. In the previous lesson, we learned about why the atmosphere moves and how that results in three distinct circulation cells in each hemisphere. In the process of moving all that air around, these winds also blow over the surface of the ocean and are largely responsible for flow in the mixed layer of the ocean. In the figure below, you should be able to recognise the arrows representing the westerly winds and [trade winds](#) that exist in both hemispheres as a result of the Hadley and Polar cells.



Prevailing winds blow over the ocean, leading to the formation (among other things) of the ocean gyres.

As the [trade winds](#) blow towards the west at the equator and the westerlies blow towards the east (yes, I know it's confusing) at the high latitudes (50-60°), the winds push water around each ocean basin, creating persistent currents. In the North Atlantic (between North America and Europe/Africa), we have the very famous Gulf Stream originating from the North Equatorial Current. The Gulf Stream then separates from North America and becomes the North Atlantic Drift before returning south towards the equator as the cooler Canary Current (though some water also flows north into the Norwegian Current). In the figure below you can see how, much like in the atmosphere, these currents are acting to transport heat away from the equator.



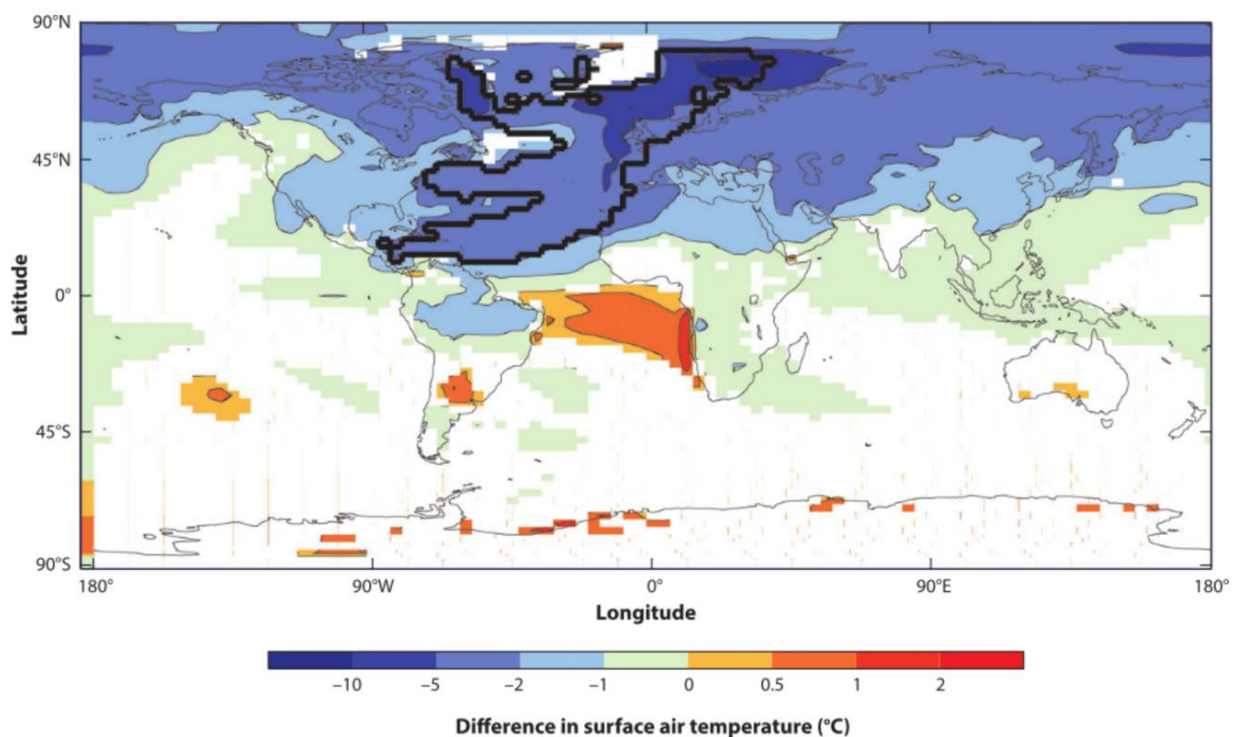
One very unique exception is the Antarctic Circumpolar Current (or ACC for short). You may notice that the stretch of ocean around Antarctica is the only place on Earth where you can go around the world without touching land. It also happens to be at the latitude where the westerly winds blow in the Southern Hemisphere. The result of this is that the ACC, like the winds that drive it, is the most powerful ocean current on Earth and along with the westerly winds, acts to

thermally isolate Antarctica from the far warmer midlatitude air and ocean temperatures, helping keep Antarctica in a [glacial](#) state.

The ocean and regional climate

As you might expect, the combination of heat transport by the ocean and atmosphere towards the poles means that there are places at higher latitudes that would be far less hospitable if these ocean and atmospheric circulations didn't exist. If we take a look at the Gulf Stream in the figure below, for example, you can see how it creates the North Atlantic Drift - supplying that warm water to the west coast of Europe. This, in combination with the prevailing westerly winds, means that Europe is a relatively warm place to live compared to the east coast of Canada (which sits at the same latitude). You'll be far less likely to hit the beach in icy Newfoundland than say, the south of England (at least in Summer, anyway).

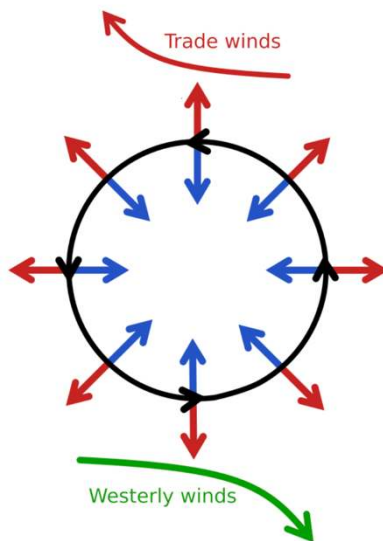
Apart from the wind-driven poleward heat transport, we will learn later about ocean circulation driven by density: this too works in such a way to transport heat from the [tropics](#) to the poles. More on that later. To examine the impact of this poleward transport in the Atlantic on European climate, certain groups have done experiments examining what would happen if this circulation were to shut down (see the figure below). Perhaps unsurprisingly, when you shut down the primary ocean current that transports heat from the equator to the poles, the high latitude regions become quite a bit cooler.



The difference in surface air temperature between a world in which the Atlantic circulation is off vs. on. (Figure from [Vellinga & Wood, 2007](#) and [Palter, 2015](#))

Returning home to Australia, we also have a boundary current off our east coast, made famous by the movie 'Finding Nemo'. Like other western boundary currents, the East Australian Current (EAC) brings warm tropical waters south and allows for the great swimming conditions at Sydney's beaches.

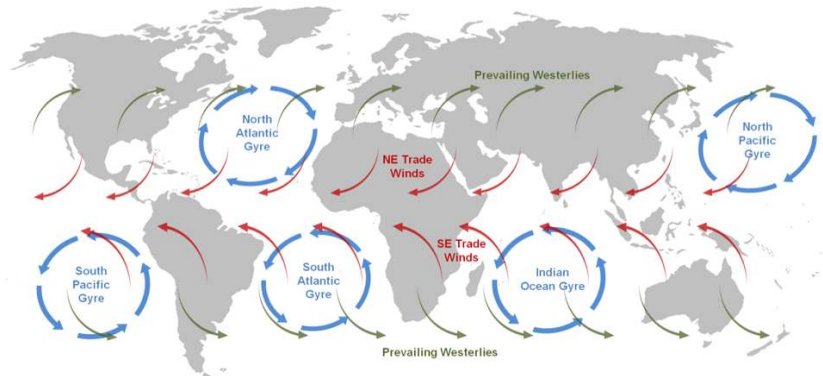
Gyres and garbage



Another interesting feature of the surface circulation are the ocean gyres. You can think of them as giant whirlpools in the middle of each basin formed due to the prevailing trade and westerly winds in each hemisphere. But why do they form as these whirlpools? It is due to a balance of forces (don't worry, this won't be assessed).

In the figure on the left we have an idealised ocean gyre in the southern hemisphere (remember, in the SH, the westerlies are to the south of the [trade winds](#)). The blue arrows represent the [Coriolis force](#) which deflects currents towards the left in the SH but in order for the current to maintain balance, something has to push back. In these gyres, the [Coriolis force](#) is constantly pushing the water towards the centre of the gyre, creating a sort of 'hill' of water - due to gravity this hill wants to flatten itself out. The red arrows are therefore the **pressure-gradient force** - the mound of seawater trying to flow out of the gyre.

When these forces balance, we have something called geostrophic flow (you don't need to remember that) and a stable ocean gyre! These gyres have a huge impact in transporting heat from the [tropics](#) to the poles via the Western Boundary Currents. Once in the higher latitudes, the water releases this heat to the atmosphere.



(Left) The forces acting on a gyre. The red arrows represent the pressure [gradient](#) force while the blue represent the [Coriolis force](#) (Adapted from image by [Ogre](#)). (Right) the five major ocean gyres.

So apart from ocean heat transport, why do we care about gyres? Well, one reason is the great ocean garbage patches. Remember how gyres are constantly having water piled into their centre? This is how they get their nickname of the ocean's vacuum cleaners because a lot of our garbage invariably ends up there. If you'd like to see how this works, click on the image below. It leads to a website created by a former CCRC researcher and allows you to dump digital rubber duckies anywhere in the ocean. Watch how the particles spread and see if you can recognise some of the currents you've learned about in this lesson.

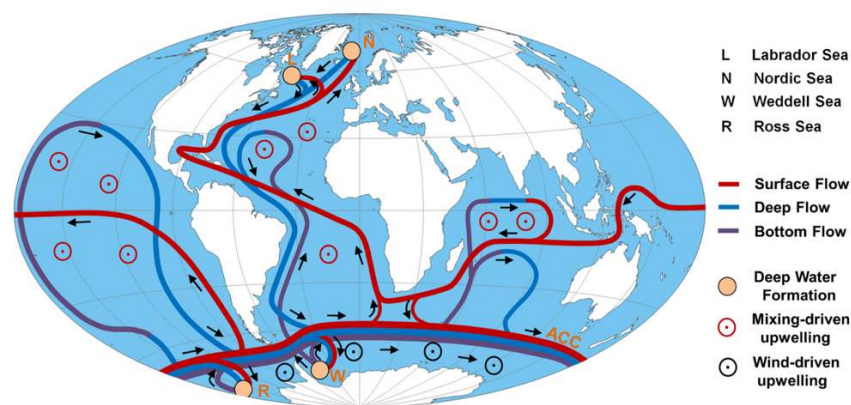
Deep ocean circulation

For the last part of this lesson, we'll be talking about the deep circulation of the ocean and how it moves below the surface mixed layer. This is where a very distinctive feature of the ocean comes into play. If you think back to the previous lesson, you'll remember that the atmosphere is heated from **below**. This makes it inherently unstable; hot air wants to rise, so the heating of the atmosphere ensures constant [convection](#) and motion. Vertical motion in the ocean is also density driven but the way in which it is heated is the exact opposite to the atmosphere - It is only heated from **above** (we're ignoring the very small amount of geothermal heating). The result is that the more you heat the surface of the ocean, the warmer it gets, the less likely it is to move vertically - heating **increases** [stratification](#).

This is why in the ocean, instead of having lighter (warm) water rise, vertical motion is driven by the sinking of cold and/or salty dense water. Once surface currents reach the poles, they have lost most of their heat to the atmosphere and when the ocean gets cold enough, some of it will begin to form into sea ice. As ice is frozen fresh water, the salt from the seawater is left in the surrounding water, making it denser. As a result, in certain regions around Antarctica and in the North Atlantic ocean, very cold, salty water will become dense enough to sink down into the ocean. Much like in the atmosphere, this process is called [convection](#). Unlike surface ocean currents which can flow at up to 1.5 m/s (Gulf Stream), vertical velocities in regions of [convection](#) are far slower, on the order of 1.5-10 m/day.

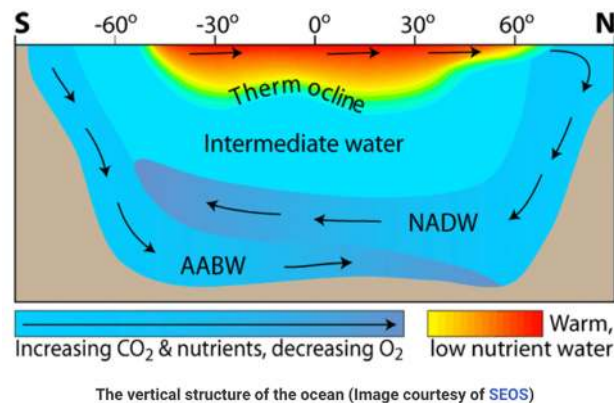
The figure below shows an example of how this occurs around Antarctica. Warm surface waters come in from the equator and cool, releasing their heat to the atmosphere. Once sufficiently cold, the formation of sea ice occurs, resulting in the surrounding water getting saltier. This water, now much denser due to the low temperature and high salinity, sinks down to the bottom of the ocean and flows back towards the equator. In Antarctica, we (creatively) call this Antarctic Bottom Water (AABW) while in the North Atlantic, this is called North Atlantic Deep Water (NADW), as unlike its Antarctic counterpart, it does not quite sink to the bottom of the ocean basins.

In the figure below we present an approximation of the world's combined surface and deep ocean (driven by density forces) circulation - the ocean's so-called 'overturning' circulation. In the North Atlantic, you can see the two NADW formation sites in both the Labrador and Norwegian seas, while in the Southern hemisphere AABW forms in the Ross and Weddell seas. These deep currents move slowly north and through mixing processes and upwelling, the water eventually resurfaces. The timescales involved in the deep circulation of the ocean mean that it can be hundreds or even several thousand years before a [parcel](#) of water sees the surface of the ocean again.



(Figure adapted from Fig.1, Kuhlbrodt et al., 2007)

This global circulation of warm surface water flowing poleward followed by cold salty deep water returning to low latitudes is what we call the thermohaline circulation. It is named this due to the fact that it is a density driven circulation which is defined by the temperature (thermo) and salinity (haline) of the water. Finally, if we take a cross section of the ocean, the figure below shows roughly what it looks like. Warm surface waters, overlying slightly cooler 'intermediate' waters (we're not concerned with these in this lesson) then finally, North Atlantic Deep Water and Antarctic Bottom Water - the coldest and densest water on Earth.

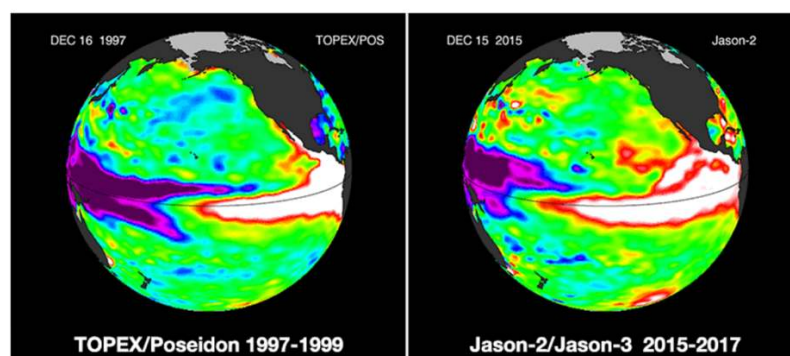


3 - Modes of variability

What is a mode of climate variability?

If the rigid circulation cells and ocean currents of the previous two lessons can be thought of as 'climate', [modes of variability](#) can be thought of as a very particular kind of weather. These phenomena have particular characteristics in space and time that are always roughly the same and are usually cyclical in nature (for example, patterns in sea surface height during an El Niño event - see figure below).

If this all sounds a bit vague, that's because it is. Methods of defining a mode of [variability](#) are varied. For example, the Southern Annular Mode (SAM – which we will discuss later) can be thought of as simply representing the pressure difference between 40°S and 60°S or, for you mathematical types, the leading empirical orthogonal function of 850 hPa atmospheric pressure [anomaly](#). The key here is that the SAM appears to be a persistent feature of the [climate system](#) and not simply a one-off weather event. Let's now take a look at three [modes of variability](#), what they are and the effect they have on our climate.



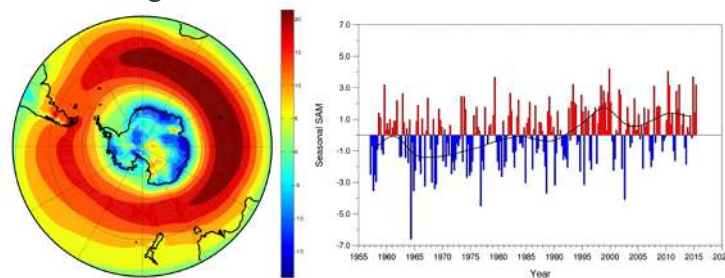
[Anomalies](#) of sea surface height (red is positive, blue is negative) in the Pacific ocean during the 1998 and 2016 [El Niño](#) events (animation courtesy [NASA](#))

Examples of modes of variability

The Southern Annular Mode (SAM)

The Southern Annular Mode (SAM) is the leading feature of atmospheric [variability](#) in the southern hemisphere. As previously mentioned, it can simply be thought of as the average pressure difference between the subpolar (40°S) and polar (60°S) latitudes. What this means in practice is that the SAM is describing the position and strength of the strong westerly winds that circle Antarctica (see figure below left).

When the SAM is positive, the winds are stronger and move closer to Antarctica, when the SAM is negative, the opposite is true. A positive SAM means less cold, low pressure air makes it north to the coast of Australia and results in less rain during the autumn and winter months. A negative SAM meanwhile means more cold fronts penetrating to lower latitudes and usually means more rainfall and stronger storms for southern Australia.

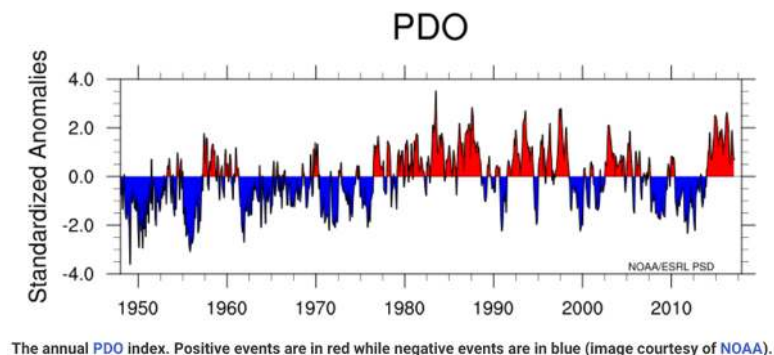


(Left) Annual mean zonal (west-to-east) wind speed in ms^{-1} (850 hPa) in the Southern hemisphere. (Right) Seasonal SAM index - the black line represents a decadal average (Image courtesy of [G. Marshall](#))

Unlike the other [modes of variability](#) we'll look at in this lesson, human influence on the SAM has been observable for several decades now. The destruction of stratospheric [ozone](#) (the [ozone hole](#)) and later warming due to [anthropogenic](#) greenhouse gases has led to a clear positive trend in the SAM since the 1970's. This has meant that, particularly in spring and summer, the winds have increased in strength and moved closer to Antarctica. As you might expect, a more positive SAM means less rain in Southern Australia and has been linked to the 'big dry' drought during 1997 - 2010.

The Pacific Decadal Oscillation (PDO)

The [Pacific Decadal Oscillation](#) is a fluctuation of sea surface temperatures in the northern and equatorial Pacific ocean basin which cycles through roughly once every 20-30 years (see figure directly below). It has two phases: a positive, warm phase and a negative, cool phase (see figures below left and right).



The annual [PDO](#) index. Positive events are in red while negative events are in blue (image courtesy of NOAA).

During the positive phase, a horseshoe of warmer water in the eastern Pacific surrounds a blob of cooler water (figure below left) and creates an [El Niño](#)-like warm patch in the eastern equatorial Pacific. The opposite is true for the cooler negative phase, where a horseshoe of cooler water surrounds a warm blob in the North Pacific (figure below right). During the warm phase, there tend to be more [El Niño](#) events which are more intense (the record 1997/98 and 2015/16 [El Niño](#) events were accompanied by a positive [PDO](#)) while the opposite is true for the negative [PDO](#).

The [impacts of the PDO](#) are similar to [El Niño](#) (which we will learn about in the next section) but less intense. A positive [PDO](#) is generally associated with drier weather in Australia, Korea, Japan, eastern Russia, northern North America and parts of Central America.

The El Niño Southern Oscillation (ENSO)

[ENSO](#) or [El Niño](#), as it is more commonly known, is arguably the most important mode of [variability](#) on Earth and its impacts are global. Much like the [PDO](#), you can find [ENSO](#) in the Pacific Ocean roughly centred on the equator. [ENSO](#), much like the SAM can be defined in a multitude of different ways, but to keep it simple, think of it as the pressure difference between Darwin in northern Australia and Tahiti in the eastern tropical Pacific (the index that measures this pressure difference is called the Southern Oscillation Index). As you can see in the figure below, much like the SAM and [PDO](#), [ENSO](#) displays an oscillatory pattern. It occurs once every 3-5 years and generally persists from anywhere between 6 months to 2 years. Before we go any further, watch the video below for a quick introduction, then we'll go into more detail.

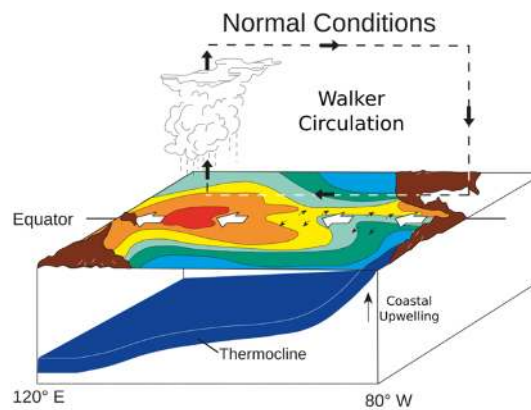
The system of air-sea interaction in [ENSO](#) is a fascinating example of positive [feedbacks](#) in the [climate system](#) and will be the focus of the remainder of the lesson. Under neutral conditions (see the figure below) the [trade winds](#) blow across the equatorial Pacific from east to west. This constant flow of water means that water is piling up on the western side of the basin, making the sea level there higher than on the eastern side. But all the warm water being pushed to the west needs to be replaced. This results in a lot of cold, nutrient-rich water being drawn up from the deep ocean on the eastern side of the Pacific.

With all this warm water on the western side, you get [convection](#) in the atmosphere, making northern Australia a very warm, tropical environment. Meanwhile, this rising air needs to go somewhere. As the water rains out and it cools, the air flows back towards the east, coming down again on the eastern side of the basin. We call this atmospheric cell the Walker circulation.

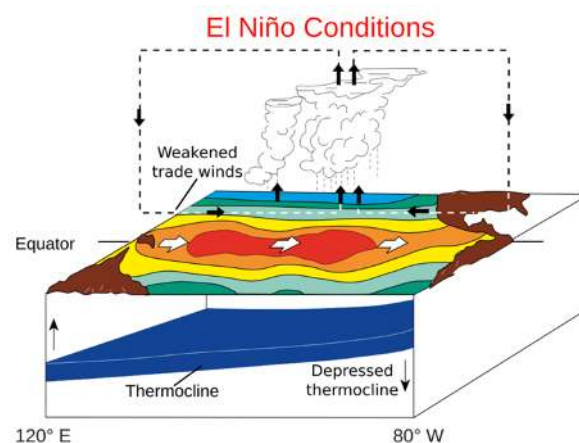
Do you notice how these two mechanisms re-enforce each other?

1. Wind blows water to west creating warm water pool in western Pacific
2. [Convection](#) occurs above warm pool, driving the Walker circulation
3. Walker circulation blows water to the west, re-enforcing the warm pool

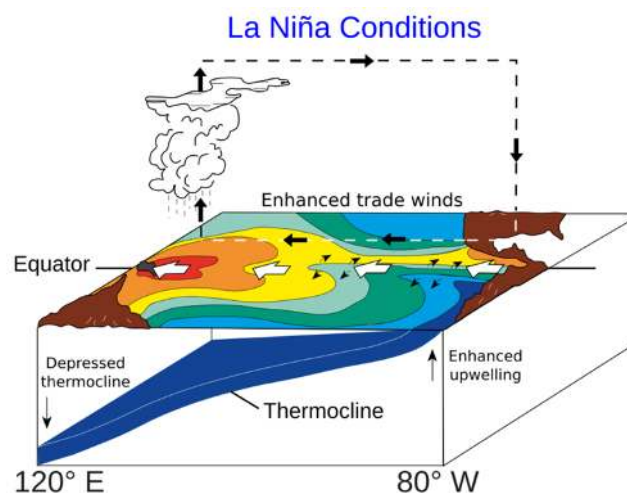
This is a classic positive [feedback](#) where both mechanisms re-enforce the other.



During the [El Niño](#) phase, the [trade winds](#) weaken and the western warm pool of water begins to expand towards the east. During this phase, rainfall which normally occurs on the western side of the basin shifts east along with the warm pool. Most of this rainfall now falls over the ocean. Meanwhile, in the ocean, the warmer waters off the coast of South America depress the [thermocline](#), preventing the upwelling of cold, nutrient-rich water. This greatly harms the fishing industry in countries such as Peru and Chile.



On the other hand, you can also have a [La Niña](#) event. This is effectively just a supercharged version of normal conditions, with enhanced [trade winds](#), even more warm water in the western Pacific and even more rainfall in places like Indonesia, Papua New Guinea, and Australia.



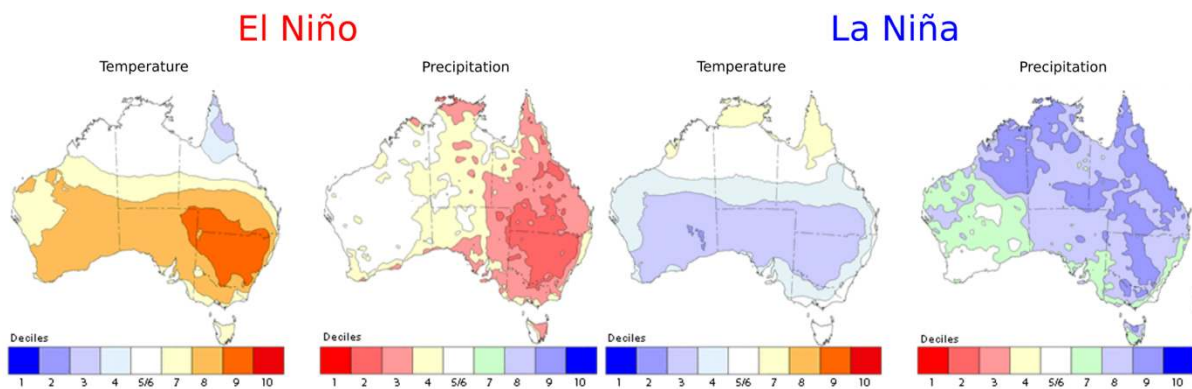
The impacts of ENSO

Now that we know how [El Niño](#) works, let's take a quick look at some of its impacts on regional climate in different parts of the world. You should remember that during an [El Niño](#), the [convection](#) and precipitation normally found in the western Pacific moves east. This is reflected in the summertime climate of Australia and South East Asia (Figure below, left) which become hotter and drier (hotter because evaporation of water isn't removing heat from the surface - but you don't need to know this). Meanwhile, in the eastern Pacific, a suppression of upwelling and warmer than normal surface waters mean their summertime climate is also warmer than normal and also much wetter. Something similar is seen in the winter months, with drier than normal conditions in the western Pacific and warmer, wetter conditions in the east.

Of course, the opposite is broadly true during a [La Niña](#), where the western Pacific gets wetter while the eastern Pacific gets cooler and drier.

Australia is particularly strongly affected by [ENSO](#) events. As can be seen in the figures below on the left, [El Niño](#) events result in considerably higher than average temperatures in the south of the country during winter/ spring as well as greatly reduced precipitation on the east coast. Once again, the effect of [La Niña](#) is roughly the opposite, with lower temperatures and increased rainfall (figure below right) during the winter/ spring months when their effect is strongest.

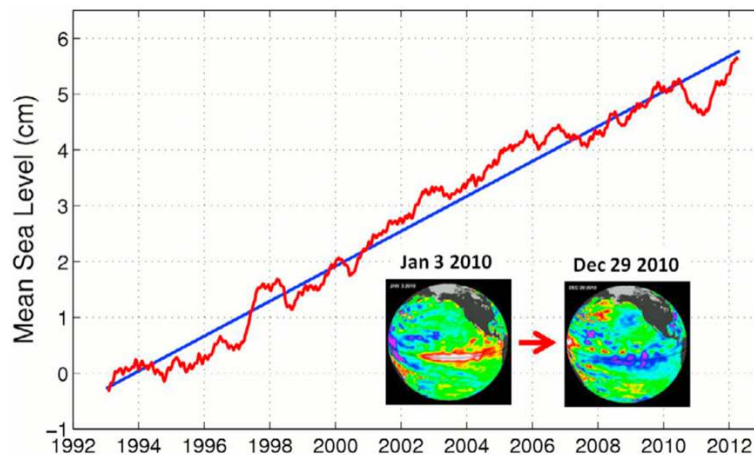
The combinations of these changes in temperature and precipitation can lead to drought or flooding, crop failures, bushfires and coral bleaching just to name a few impacts.



Daytime temperatures and daily precipitation in winter and spring for [El Niño](#) and [La Niña](#) events. Units are in [Deciles](#). All figures courtesy of the [BoM](#).

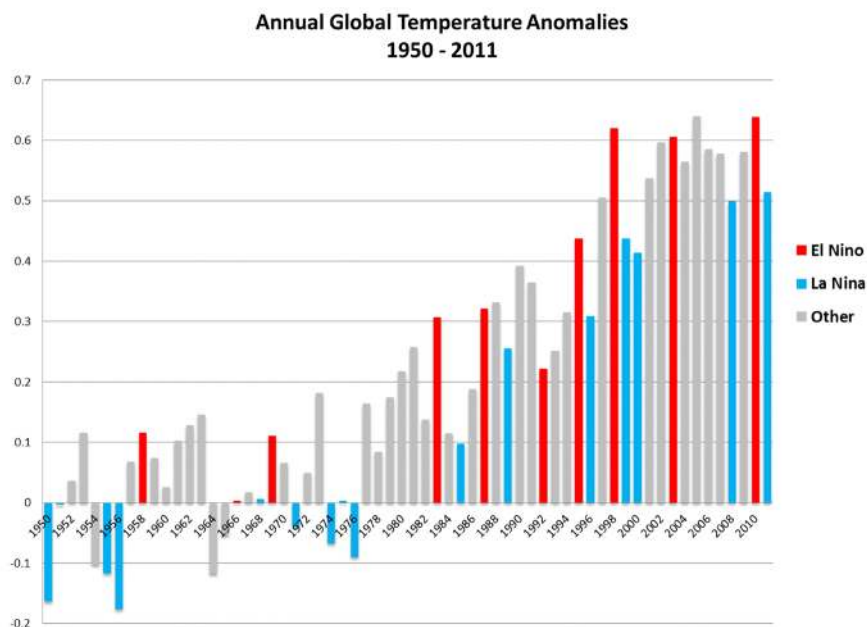
Impacts of ENSO continued

So what happens when a particularly big [ENSO](#) event occurs? In 2010-2011, we experienced a particularly intense [La Niña](#) which happened to coincide with favourable IPO and SAM conditions, resulting in considerable flooding in Queensland - 3/4 of the state was declared a disaster area and 38 people lost their lives. The financial cost was estimated at over \$2 Billion. The most shocking part of this, however, is that so much water fell on land (and in Australia in particular) that for 2011 the global mean sea level actually dropped (see figure below, look at 2010-2011). If you want to read more about this, check out [this article](#).



Global mean sea level with blue linear trend (Boening et al., 2012)

What about temperature? Well, considering that about 93% of global warming is being absorbed by our oceans rather than remaining in the atmosphere, any change in the way that the ocean and atmosphere exchange heat should have a noticeable impact on global surface air temperatures. This proves to be true - during [El Niño](#) years, the warm Pacific ocean releases more heat to the atmosphere, making [El Niño](#) years warmer than average (see figure below). The opposite is true for [La Niña](#) years, during which the tropical Pacific is cooler than usual and the ocean actually absorbs more heat than usual. If you'd like to know more about this and how [ENSO](#) contributed to the record temperatures we've seen in the last few years, [this article](#) is well worth a read.



Annual global temperature anomalies for El Niño, La Niña and neutral years (NOAA state of the climate report, 2011)